The Claimed Complex Structure on S^6
Validate, Repair, or Disprove

Working mathematical notebook following the three-stream proof audit

26 August 2026

This workbench has migrated.

Further mathematical work and the consolidated source tree are maintained at https://www.overleaf.com/read/rtmyqxyrzprn#fa24eb. This preserved project remains available as an earlier checkpoint.

Abstract

This paper validates, repairs, or attempts to disprove a reconstructed manuscript claiming a compact complex threefold diffeomorphic to S^6 and of algebraic dimension one. In its present form the workbench gives complete proofs of the period-family descent, toric and finite fillings, global analytic gluing, integral nearby-cycle specialization, the global Leray page and three primitive transgressions, the based van Kampen presentation, sphere recognition, and the very-general-fibre Néron–Severi and meromorphic-descent calculation. It therefore verifies for the constructed threefold

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2, \qquad \mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1.$$

The same verified object satisfies the stated hypotheses of Campana–Demailly–Peternell Theorem 2.2 and contradicts its conclusions, so the workbench records a counterexample to that theorem as stated as well as two exact defects in its printed dependency chain. The separate cellular-collapse/Mayer–Vietoris comparison remains open as an alternate certificate. The candidate's Proposition 10.8 is now also verified: the structure-sheaf direct images, degree-one base change, Picard line, exact finite normal characters, and uniform period/countability argument are reconstructed in full without using that open collapse route. Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff and provisionally agreed on the non-collapse route; the later Codex proofs here are separate. Throughout, **[VERIFIED]** means that a complete auditable proof is written in this document, not that specialist replication or community acceptance has already occurred.

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1 Results and evidence boundary

The target is a reconstructed manuscript claiming a compact complex threefold X and holomorphic map $f : X \to \mathbb{P}^1$ with

$$X \cong_{\text{diff}} S^6, \qquad a(X) = 1, \qquad \langle c_3(T_X), [X] \rangle = 2.$$

As of 26 August 2026, the workbench records the following mathematical status. Here **[VERIFIED]** means that a complete proof is written in this workbench; it is not a proxy for a model verdict or for community acceptance.

1. **[VERIFIED]** The modular period family, toric cusp filling, finite logarithmic fillings, punctured-overlap biholomorphisms, and global gluing produce a connected compact smooth complex threefold and a proper holomorphic map
$$f : X \longrightarrow \mathbb{P}^1$$

with the asserted three singular fibres; see theorems 6.1 and 6.3 to 6.7.

2. **[VERIFIED]** The monodromy, cusp invariant lattices, global coinvariants, finite-filling stalk maps, terminal group presentation, and terminal degree-two lattice are proved over $\mathbb{Z}$ in theorems 5.1 to 5.6.

3. **[VERIFIED]** The intrinsic integral nearby-cycle specialization at the cusp is injective with saturated image
$$\operatorname{im}(\mathrm{sp}^q) = H^q(F;\mathbb{Z})^{T_0} \qquad (0 \le q \le 4),$$
and the finite stalks and parabolic sheaf calculation give the complete global Leray page. The canonical truncation triangle identifies the explicit Cech discrepancy cocycle with Leray d_2, and multiplicativity then gives
$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$
Consequently
$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0;$$
see theorems 7.1 to 7.5.

4. **[VERIFIED]** With zero-section whiskers and the fixed clockwise meridians, the actual overlap inclusions give
$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$
Together with the preceding integral cohomology this proves
$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2;$$
see theorems 7.6 and 7.7.

5. **[VERIFIED]** The very-general fibre has
$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad \eta^2 = 12, \qquad a(F_b) = 0,$$
and the repaired meromorphic-descent argument gives
$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$
with f the algebraic reduction; see theorems 8.2 and 8.4.

6. **[REFUTED]** The p. 692 sheaf-theoretic inference of Campana–Demailly–Peternell is false in the stated generality. If $\mathscr{K}_S = \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S)$, then
$$0 \longrightarrow N^*_{S/X} \longrightarrow \mathscr{K}_S \longrightarrow \operatorname{tors}\Omega^1_S \longrightarrow 0,$$
and the two printed vanishings do not control sections mapping to the conductor-supported quotient. The proper nodal-cubic-times- $\mathbb{P}^1$ family in theorem 11.6 is an explicit counterexample to that inference.

7. **[VERIFIED]** The conductor section, nonnormal Cartier-divisor duality, flatness, and top-degree analytic base change are complete in theorems 11.5 and 11.7 to 11.9. Combining them with the now-verified construction and $H^2(X;\mathbb{Z}) = 0$ gives
$$R^2 f_*(T_X \otimes L) \neq 0 \qquad \text{for every } L \in \mathrm{Pic}(X).$$

8. **[REFUTED]** The separate split-extension calculation imported from CDP98 replaces monodromy coinvariants by the entire fibre homology. The correct formula is

$$(K \rtimes Q)^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q.$$

9. **[VERIFIED]** The constructed X satisfies the stated hypotheses of CDP20 Theorem 2.2 but has both $c_3(X) = 2 > 0$ and $a(X) = 1$. Thus it is a counterexample to that theorem and its ensuing S^6 corollary as stated. This theorem-level conclusion uses the complete construction, topology, and algebraic-reduction proofs above; it is not inferred merely from the two printed proof-route defects.

10. **[VERIFIED]** The candidate's Proposition 10.8 conclusion holds. The workbench proves

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \mathrm{Pic}(X) \simeq \mathbb{C},$$

degree-one base change at every scheme-theoretic fibre, and $\mathrm{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. The smooth-fibre period condition is exactly $c\tau + d\mu - b = 0$; hence every parameter outside $\mathbb{R}$ and a countable set gives one line bundle non-torsion on every full fibre and on both S_j. The cusp extension uses intrinsic nearby cycles rather than the open collapse comparison; see theorems 9.4 and 9.6.

11. **[OPEN]** The separate cellular-collapse and Mayer–Vietoris comparison route remains under audit: C1–C3, MV1, and the comparison part of L1 are not closed. These are alternate certificates. They are not hypotheses of the verified intrinsic nearby-cycle/Leray proof above.

Independent review checkpoint. Reviewer B independently examined the reconstruction existing up to that report's evidentiary cutoff and agreed that the intrinsic nearby-cycle, Leray, and based van Kampen route does not depend on the open collapse route. Its disposition was provisionally favourable with focused verification requests. The later Codex proofs recorded here are separate work: they are not retroactively attributed to Reviewer B, and its report remains useful as an independent comparison against the argument it actually saw.

Status language. **[VERIFIED]** means proved here or reproduced by an explicit certificate. **[CONDITIONAL]** means a proved implication with every unverified hypothesis named; it never means doubtful or politely false. **[OPEN]** means that no complete proof or counterexample is recorded. **[REFUTED]** means an exact assertion has a complete contradiction or counterexample. The status applies to the immediately following assertion, not to an entire paper by contagion.

2 Mathematical results so far

This is the short route through the current mathematics. It states what is proved in this workbench, exhibits the load-bearing formulas, and separates the verified intrinsic nearby-cycle/Leray proof from the still-open cellular-collapse comparison.

1. **[VERIFIED] The analytic threefold exists in the reconstructed coordinate model.** The descended modular periods give

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix},$$

with

$$\det_{\mathbb{R}} \Pi = \mathrm{Im}\,\tau \left(\mathrm{Im}\,\beta - \frac{6(\mathrm{Im}\,\mu)^2}{\mathrm{Im}\,\tau} \right) \neq 0.$$

The periodic toric cusp action is free and properly discontinuous; its bounded-chart argument uses $|B_t\bar{\lambda}| \geq \frac{1}{2}|\bar{\lambda}|$. Exponentiation identifies the punctured toric quotient with the abelian surface family. The finite logarithmic actions are free by the invariant covector criterion, their radial stalk maps are the printed quotient maps, and the three fillings glue to a compact connected smooth complex threefold X with a proper surjective holomorphic map

$$f : X \longrightarrow \mathbb{P}^1.$$

The complete chain is theorems 6.1 and 6.3 to 6.7.

2. **[VERIFIED] The cusp fibre has the exact geometry needed in the topology and conductor arguments.** Its scheme-theoretic local equations are

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2.$$

It is reduced, irreducible, nonnormal, and normal crossings (not simple normal crossings). Its normalization is dP_6, with opposite sides of the anticanonical hexagon identified, and its double locus consists of three rational curves through two triple points.

3. **[VERIFIED] The algebraic reduction is computed directly and does not use the topology or CDP20.** For a very general smooth fibre,

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad \eta = uw + 6\gamma\delta, \qquad \eta^2 = 12.$$

The Appell–Humbert form has determinant

$$\det G_\eta = \frac{2}{3}\operatorname{Im}\tau\left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}\right) < 0,$$

so it has signature $(1,1)$ and $a(F_b) = 0$. The repaired meromorphic-descent argument then gives

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction; see theorems 8.2 and 8.4.

4. **[VERIFIED] The intrinsic nearby-cycle/Leray route proves the integral topology without the disputed cellular collapse.** For the cusp filling, the integral nearby-cycle map satisfies

$$\mathrm{sp}^r : H^r(W;\mathbb{Z}) \xrightarrow{\sim} H^r(F;\mathbb{Z})^{T_0} \qquad (0 \leq r \leq 4)$$

as saturated lattices. The global sheaf calculation gives

$$H^0(R^1) = \mathbb{Z}(12\gamma), \quad H^0(R^2) = \mathbb{Z}(2q), \quad H^0(R^3) = \mathbb{Z}(2\gamma uw).$$

The canonical truncation triangle

$$R^0f_*\mathbb{Z}_X \longrightarrow \tau_{\leq 1}Rf_*\mathbb{Z}_X \longrightarrow R^1f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0f_*\mathbb{Z}_X[1]$$

identifies the explicitly computed Cech discrepancy with Leray d_2. For $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$, hence $p = -1$, orientation of ω may be chosen so that

$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$

Therefore

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0.$$

5. **[VERIFIED] The based van Kampen maps and sphere recognition close the topology theorem.** Zero-section whiskers fix the common basepoint, the finite logarithmic overlaps impose the primitive relations with one common sign, the cusp inclusion is $\Lambda \to \bar{\Lambda}$, and the cusp meridian bounds the explicit toric zero-section disc with no translation drift. Hence

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$

Poincaré duality, universal coefficients, Hurewicz, and the smooth homotopy-sphere recognition theorem now give

$$X \cong_{\text{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = \chi_{\text{top}}(X) = 2.$$

See theorems 7.1 to 7.7.

6. **[REFUTED] The printed CDP dependency has two exact failures.** For a reduced nonnormal fibre,

$$0 \longrightarrow N^*_{S/X} \longrightarrow \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S) \longrightarrow \operatorname{tors} \Omega^1_S \longrightarrow 0,$$

so the vanishings used on p. 692 of [2] do not control the conductor-supported middle term. A proper nodal-cubic-times-$\mathbb{P}^1$ family is a counterexample to that inference. Separately, a split extension satisfies

$$(K \rtimes Q)^{\text{ab}} \simeq Q^{\text{ab}} \oplus (K^{\text{ab}})_Q,$$

not the direct sum with all of K^{ab} unless monodromy acts trivially.

7. **[VERIFIED] The constructed threefold is a theorem-level counterexample to CDP20 Theorem 2.2 as stated.** It has

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1, \qquad f : X \to \mathbb{P}^1 \text{ a holomorphic algebraic reduction,}$$

while

$$c_3(X) = 2 > 0, \qquad X \cong_{\text{diff}} S^6.$$

These are precisely the relevant hypotheses and the opposite of the published conclusions $c_3(X) \leq 0$ and $a(X) = 0$. This conclusion uses the verified candidate proof, not merely the fact that defects were found in the published proof.

8. **[VERIFIED] The Proposition 10.8 uniform non-torsion branch is complete.** The reconstructed direct-image calculation gives

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \operatorname{Pic}(X) \simeq \mathbb{C},$$

with degree-one base change on all scheme-theoretic fibres. The finite character calculation gives $\operatorname{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. For $\lambda = a\gamma + bu + cw + d\delta$, membership of its Dolbeault image in the global Picard line is exactly

$$c\tau + d\mu - b = 0.$$

Linear independence of $1, \tau, \mu$ makes the rank-two exceptional fibres countable, while rank-one fibres contribute bad parameters only in $\mathbb{R}$. The three full special fibres and the reductions S_j each add only countably many parameters. Thus every $z \in \mathbb{C} \setminus (\mathbb{R} \cup \text{a countable set})$ supplies one L_z non-torsion on every full fibre and on both S_j. The cusp step is a Codex repair using intrinsic nearby cycles, not the open cellular collapse.

9. **[OPEN] The cellular-collapse/Mayer–Vietoris route remains an independent comparison, not a gap in the proof above.** The continuity of the separate equivariant collapse, its primitive integral matrices, equality with intrinsic specialization, and the oriented Mayer–Vietoris comparison are the active C1–C3, MV1, and L1 tasks. Closing them would produce a second topological certificate and validate the candidate's stronger claims about its explicit collapse map.

Exact-computation certificates. The retained exact-arithmetic harnesses passed all 28 full-manuscript checks and all 38 independently implemented core checks, with no failures. They cover finite and cusp monodromy, invariant and coinvariant lattices in exterior degrees 0 through 4, Smith forms and saturation, finite-quotient norm operators, the relation-group determinant, specialization indices, three twisting parameters, period-form identities, and the corrected existential form of Remark 7.28. The geometric realizations and comparison maps are not inferred from those executions; their proofs are written separately in the sections cited above.

Independent review checkpoint. Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff. It withdrew the earlier topology-based rejection, distinguished the intrinsic specialization sp^q from the open collapse pullback c^*, and found no load-bearing defect in the non-collapse nearby-cycle/Leray and based van Kampen route. Its provisionally favourable disposition and focused verification requests are useful independent evidence about the argument it saw. The later Codex proofs in this workbench are separate and should not be attributed to that review.

Current evidence boundary. No mathematical gap is presently recorded in the central non-collapse proof of the construction, integral topology, sphere recognition, or algebraic dimension. The labels above mean that complete arguments are present here for audit; they do not replace specialist replication. The still-open work is exact and narrower: the alternate collapse/Mayer–Vietoris comparison, the uncomputed value $h^{1,2}$, and any independent repair or corrected replacement of CDP20 Theorem 2.2.

3 Methodology

The unit of audit is an exact mathematical assertion, not a paragraph's plausibility. Each assertion is recorded with its domain, codomain, coefficient ring, variance convention, and dependencies. The four outcomes used in this paper are literal:

Validated A complete proof from the stated hypotheses is given or a precisely cited theorem is applied after every hypothesis is checked.

Repaired The printed assertion fails as stated, but a corrected statement is proved and every downstream use is recomputed.

Disproved A complete contradiction or counterexample is given to the exact assertion. A defect in a proof is not, by itself, a disproof of its theorem.

Open No complete proof or counterexample is presently recorded. This is not a euphemism for false.

Computations are converted into checkable certificates: displayed matrices, integer row or column relations, determinant ideals, primitive kernels, or explicit substitutions. A software pass is corroboration, not a proof. Likewise, an algebraically correct matrix calculation does not identify that matrix with a geometric specialization, attaching, or boundary map; that identification receives its own statement and proof obligation.

Relevant human literature is searched proactively, not only when a remembered fact seems doubtful. Results are cited at the point of use with theorem, section, or page locators and with their hypotheses intact. Public working LaTeX editions such as the maintained SGA English source [25] may be used as searchable reading layers, but the published SGA volumes [21, 22] remain canonical. Every reference in the compiled paper is resolvable without access to the editor's computer. The citation and tool-disclosure policy follows the Leiden Declaration [1]; neither a model-generated review nor an internal audit note substitutes for a human mathematical source.

4 Relation to existing literature and exact scope

The theorem-level conflict is with Campana–Demailly–Peternell [2, Theorem 2.2 and Corollary 2.3, p. 680]. Their hypotheses match the candidate's asserted $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction. Their conclusion gives $c_3(X) \leq 0$ and, for a threefold homeomorphic to S^6, algebraic dimension zero. The candidate instead asserts $c_3(X) = 2$ and $a(X) = 1$. The conflict is therefore exact if the candidate assertions are true; it is not resolved by observing that the candidate proof does not invoke [2].

The first local issue isolated below occurs in the opening reduction of [2, §7, p. 692]. The explicit nodal-cubic family in theorem 11.6 refutes that reduction as a formal sheaf-theoretic inference. A second issue is the replacement of monodromy coinvariants by the full fibre homology in [3, Lemma 3.2, pp. 84–85]; CDP20 later imports the resulting rank argument. Either defect might in principle admit a different repair under the global hypotheses. No such repair is presently known here.

For the singular fibre, analytic duality is checked in two independent ways. The proof in theorem 11.7 derives the result from smooth Serre duality [8, Theorem 9.37]; Ramis–Ruget's dualizing-complex theorem provides the singular analytic comparison [4, Theorems 1–2, pp. 81–84, and Proposition 2, p. 91]. Relative duality for a proper analytic morphism is also cited in its precise form as Ramis–Ruget–Verdier, Invent. Math. 13 (1971), 261–283 [5]. Analytic cohomology and base change is applied in the form of Bănică–Stănăşilă, Chapter III, §3, Theorem 3.4 and Corollary 3.10, together with Grauert's local-freeness/continuity theorem in Chapter III, §4, Theorem 4.12 [6]. These locators cover all degrees used by the structure-sheaf direct-image proof, not only its top-degree specialization. The algebraic top-degree analogue in the Stacks Project, Tag 0EX2 [26], is a useful cross-check but is not substituted for the analytic theorem.

The algebraic nearby-cycle formalism in SGA 7 II, Exposé XIII [22], and the semistable/Raynaud framework in SGA 7 I, Exposé IX [21], do not by themselves cover every step for the present proper holomorphic map, which is not assumed algebraic. The needed analytic formalism is nevertheless available. Dimca's Definition 4.2.1 constructs nearby cycles for complex analytic varieties; Proposition 4.2.2 identifies their stalks with Milnor-fibre cohomology; Proposition 4.2.11 commutes nearby cycles with proper analytic direct image; and Theorem 2.3.26 gives proper base change, all for coefficient rings including $\mathbb{Z}$ [23]. Kashiwara–Schapira [24, §8.6] is an independent formal comparison. Thus the general integral analytic framework is verified. None of these references by itself identifies the candidate's specialization image with the full monodromy-invariant lattice. That candidate-specific conclusion, including saturation, is proved in theorem 7.1 for the reconstructed proper cusp model, using the explicit local equations and a supported Gysin calculation. Equality with the separate cellular collapse map remains the open part of obligation C3.

For compact-support Poincaré duality with local coefficients and its projection formula we use Bredon [10, Chapter VI], with Hatcher [9, §3.H] as a concise topological overview. This is the precise input behind the multiplicative identification $H^2(B, j_*\mathcal{L}) \simeq M_G$ used in the Leray calculation.

The algebraic reduction of a compact complex torus is used in the exact form given by Birkenhake–Lange [14, Chapter 2, §6], with the independent statement in Pareschi–Popa–Schnell [15, §8, p. 2437]. The toric chart facts are compared with Fulton [16]. The exact classical-topology formula $\pi_1(X_\Sigma) \cong N/N_\Sigma$ is cited from Cox–Little–Schenck [17, Theorem 12.1.10], with Danilov's original toric treatment [18, §9] as an independent comparison. All assertions specific to the locally finite infinite fan and the candidate's proper-discontinuity argument are proved directly.

Finally, the sphere-recognition endpoint uses Hurewicz and homological Whitehead exactly as stated by Hatcher [9, Theorem 4.32 and Corollary 4.33], Smale's generalized Poincaré theorem [11, Theorem A], and the computation of the group of homotopy 6-spheres by Kervaire–Milnor [12, opening table]. Those results close the recognition step after the candidate's fundamental group and integral homology are established in theorems 7.5 and 7.6; they are not used to establish those inputs.

5 Exact integral algebra of the printed monodromy data

This section replaces opaque computational status reports by certificates that can be checked by hand. Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle.$$

Vectors are columns in the displayed ordered bases. The printed monodromy on V is

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.1}$$

Put $T_0 = (T_1 T_2)^{-1}$ and let $A_j = (T_j^{-1})^t$, $M_0 = (T_0^{-1})^t$ act on Λ.

Proposition 5.1 (monodromy certificate)**.** ***[VERIFIED]*** *Direct integral multiplication gives*

$$T_1^3 = I, \quad T_2^4 = I, \quad T_0 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (T_0 - I)^2 = 0,$$

and

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

with $A_1 A_2 M_0 = I$.

Proof. Multiplying the first two matrices in (3.1) gives

$$T_1 T_2 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its inverse is the displayed T_0. Thus $T_0 - I$ sends $(a, b, c, d)^t$ to $(d, -c, 0, 0)^t$, and applying it twice gives zero. The complete order certificate is

$$T_1^2 = \begin{pmatrix} 1 & 6 & -6 & -2 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_1^2 T_1 = I,$$

$$T_2^2 = \begin{pmatrix} 1 & 6 & -6 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 = \begin{pmatrix} 1 & 0 & -6 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 T_2 = I.$$

In particular $T_1 \neq I$ and $T_2^2 \neq I$. Transposing the integral inverses gives the three displayed contragredient matrices; finally $A_1 A_2 = (M_0)^{-1}$ by the definitions, hence $A_1 A_2 M_0 = I$. □

5.1 The cusp invariant, image, and coinvariant lattices

Theorem 5.2 (all exterior-power cusp lattices)**.** ***[VERIFIED]*** *The equality*

$$\ker(M_0 - I) = \operatorname{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle \tag{3.2}$$

holds as an equality of saturated sublattices of Λ*. On* V*,*

$$V^{T_0} = \mathbb{Z}\langle \gamma, u \rangle,$$

$$(\textstyle\bigwedge^2 V)^{T_0} = \mathbb{Z}\langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \qquad (\textstyle\bigwedge^3 V)^{T_0} = \mathbb{Z}\langle \gamma uw, \gamma u\delta \rangle. \tag{3.3}$$

Every displayed invariant lattice is saturated. Consequently the invariant ranks in degrees $0, 1, 2, 3, 4$ *are* $(1, 2, 4, 2, 1)$*.*

Proof. From the displayed matrix,

$$(M_0 - I)(a, b, c, d)^t = (0, 0, b, -a)^t.$$

Its kernel is given by $a = b = 0$, and its image consists of every vector $(0, 0, b, -a)^t$; this proves (3.2). The two generators are members of the chosen basis, so the lattice is saturated.

The action of T_0 on basis vectors is

$$T_0\gamma = \gamma, \quad T_0 u = u, \quad T_0 w = w - u, \quad T_0\delta = \delta + \gamma. \tag{3.4}$$

Thus a vector $a\gamma + bu + cw + d\delta$ is fixed exactly when $c = d = 0$. For degree two, write

$$x = a\gamma u + b\gamma w + c\gamma\delta + d\, uw + e\, u\delta + f\, w\delta.$$

Expanding (3.4) and collecting coefficients gives

$$(\textstyle\bigwedge^2 T_0 - I)x = (-b - e + f)\gamma u - f\gamma w - fu\delta.$$

Hence $f = 0$ and $e = -b$, with a, b, c, d arbitrary, which is exactly the middle lattice in (3.3). These are integral equations with a unit coefficient, so their solution lattice is saturated.

In the ordered basis $(\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$,

$$(\textstyle\bigwedge^3 T_0 - I)(a, b, c, d)^t = (d, -c, 0, 0)^t.$$

The fixed lattice is therefore generated by the first two basis vectors and is saturated. Degrees zero and four are fixed because $\det T_0 = 1$. □

5.2 Global coinvariants

Let $G = \langle T_1, T_2 \rangle$ and, for a G-lattice L, write $L_G = L/((T_1 - I)L + (T_2 - I)L)$.

Theorem 5.3 (coinvariant certificates)**.** ***[VERIFIED]*** *The common invariant lattices are*

$$V^G = \mathbb{Z}\gamma, \qquad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \qquad (\textstyle\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \qquad (\textstyle\bigwedge^4 V)^G = \mathbb{Z}\,\mathrm{vol},$$

where $q = uw + 6\gamma\delta$ *and* $\mathrm{vol} = \gamma uw\delta$*. There are torsion-free identifications*

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad (\textstyle\bigwedge^3 V)_G = \mathbb{Z}[uw\delta], \tag{3.5}$$

and $[uw] = 6[\gamma\delta]$ *in degree two.*

Proof. Solving $(T_1 - I)v = 0$ and $(T_2 - I)v = 0$ in (3.1) gives

$$V^{T_1} = \langle \gamma, 2u + w + 3\delta \rangle, \qquad V^{T_2} = \langle \gamma, u + w + 2\delta \rangle.$$

If a vector belongs to both lattices, comparison of its u- and w-coordinates gives simultaneously $2r = t$ and $r = t$ for the two non-γ coefficients; hence $r = t = 0$. Thus $V^G = \mathbb{Z}\gamma$.

The higher common invariants follow from the same integral nullspace calculation. In the ordered bases

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta), \qquad (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta),$$

the simultaneous equations $(\bigwedge^k T_1 - I)x = (\bigwedge^k T_2 - I)x = 0$ are respectively equivalent to

$$x_1 = x_2 = x_5 = x_6 = 0, \qquad x_3 = 6x_4,$$

and

$$x_2 = x_3 = x_4 = 0.$$

Thus the integral solution lattices are generated by $uw + 6\gamma\delta = q$ and γuw. Both generators are primitive. Finally $\det T_1 = \det T_2 = 1$, so the common degree-four invariant lattice is $\mathbb{Z}\,\mathrm{vol}$.

In V_G, columns of $T_1 - I$ and $T_2 - I$ give, among others,

$$u + w = 0, \qquad u - 3\gamma = 0, \qquad 2\gamma + u + w = 0, \qquad 2u + w = 0.$$

The first two relations give $w = -u$ and $u = 3\gamma$; the last two then give $2\gamma = 0$ and $3\gamma = 0$. Hence $\gamma = u = w = 0$. Every column has zero δ-coordinate, so no relation involves δ. This proves $V_G = \mathbb{Z}[\delta]$, including absence of torsion.

For degree two use the ordered basis

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta).$$

In this basis the twelve columns of the relation map are

$$B_2 = \left[\begin{array}{rrrrrr|rrrrrr} -2 & 1 & 1 & -6 & 2 & -8 & -1 & -1 & 1 & -6 & 6 & -3 \\ -1 & -1 & 1 & -6 & 2 & -6 & 1 & -1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -6 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 \end{array}\right], \tag{3.6a}$$

where the two blocks are $\bigwedge^2 T_1 - I$ and $\bigwedge^2 T_2 - I$. Its column lattice is

$$\mathrm{im}[\,\textstyle\bigwedge^2 T_1 - I \mid \bigwedge^2 T_2 - I\,] = \langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle. \tag{3.6}$$

For completeness, the third columns of the two blocks yield γu and $\gamma u + \gamma w$; their difference yields γw. After these are removed, the fifth and sixth columns give $2u\delta + w\delta$ and $u\delta + w\delta$, hence both $u\delta$ and $w\delta$; the remaining nonzero relation is $uw - 6\gamma\delta$. Conversely, substituting the twelve displayed columns of the two exterior powers shows that they all lie in the right side. After ordering the ambient basis as $(\gamma u, \gamma w, u\delta, w\delta, uw, \gamma\delta)$, the five generators have matrix $I_4 \oplus (1, -6)^t$. Its maximal minors have gcd 1, so (3.6) is saturated. The quotient is therefore freely generated by $[\gamma\delta]$, with $[uw] = 6[\gamma\delta]$.

For degree three, let $(a, b, c, d) = (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$. The full relation matrix is

$$B_3 = \left[\begin{array}{rrrr|rrrr} 0 & 0 & 1 & 2 & 0 & -1 & 0 & -3 \\ 0 & -2 & 1 & -6 & 0 & -1 & -1 & -6 \\ 0 & -1 & -1 & -6 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]. \tag{3.6b}$$

If $c_{i,r}$ denotes column r of the T_i block, then $c_{1,2} - c_{2,3} = -b$, $c_{2,3} = -b - c$, and $c_{1,3} = a + b - c$. Hence a, b, c belong to the column lattice. Conversely every displayed column lies in $\langle a, b, c \rangle$, and the fourth row is zero. The relation lattice is therefore exactly the primitive summand $\langle a, b, c \rangle$, so the quotient is $\mathbb{Z}[uw\delta]$ with no hidden torsion. □

5.3 Finite quotient fibres

Put

$$\eta_1 = 2u + w + 3\delta, \qquad \eta_2 = u + w + 2\delta, \qquad \mathbf{q} = uw + 6\gamma\delta,$$

$m_1 = 3$, $m_2 = 4$, and $v_1 = (1, 2, -4, 0)^t$, $v_2 = -(1, 3, -3, 0)^t$ in Λ.

Proposition 5.4 (finite-fibre pullback indices)**.** ***[CONDITIONAL]*** *Assume the free affine actions and finite fibres are geometrically realized as in theorem 6.5. Then*

$$S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle$$

with $H^*(A_0^{(j)}; \mathbb{Z}) = \bigwedge^* V$, the degree-$m_j$ covering $\pi_j : A_0^{(j)} \to S_j$ has injective pullback with indices, in degrees 0 through 4,

$$(1, 3, 1, 1, 3), \qquad (1, 4, 2, 2, 4). \tag{3.7}$$

Moreover

$$\pi_1^* H^1(S_1) = \langle 3\gamma, \eta_1 \rangle, \qquad \pi_2^* H^1(S_2) = \langle 4\gamma, \eta_2 \rangle,$$
$$\pi_1^* H^2(S_1) = (\textstyle\bigwedge^2 V)^{T_1}, \qquad \pi_2^* H^2(S_2) = \{a\gamma\eta_2 + b\mathbf{q} : a \equiv b \pmod 2\}. \tag{3.8}$$

The abstract group and lattice calculation below is ***[VERIFIED]****; its identification with the stalks of the proposed holomorphic family remains exactly the stated geometric hypothesis.*

Proof. The fundamental group is the extension

$$\Gamma_j = \langle \Lambda, g \mid g\lambda g^{-1} = A_j\lambda,\ g^{m_j} = v_j \rangle.$$

After abelianization,

$$H_1(S_j) = (\Lambda_{A_j} \oplus \mathbb{Z}g)/\mathbb{Z}([v_j], -m_j). \tag{3.9}$$

The matrices $A_j - I$ have Smith form $\mathrm{diag}(1, 1, 0, 0)$, so Λ_{A_j} is free of rank two. The relation in (3.9) is primitive: if a prime divides m_j, its image $[v_j]$ modulo that prime is nonzero because $\gamma((A_j - I)\Lambda) = 0$ and $\gamma(v_1) = 1$, $\gamma(v_2) = -1$. Thus $H_1(S_j)$ is free of rank two. The quotient is a closed oriented four-manifold (the linear parts have determinant one) and $e(S_j) = e(A_0^{(j)})/m_j = 0$. Poincaré duality and the universal coefficient theorem therefore give

$$H_k(S_j; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z})$$

with no torsion. Indeed, Poincaré duality gives $H_3(S_j) \simeq H^1(S_j) \simeq \mathbb{Z}^2$ and $H^3(S_j) \simeq H_1(S_j) \simeq \mathbb{Z}^2$. The universal coefficient sequence for H^3 then forces $\mathrm{Ext}(H_2(S_j), \mathbb{Z}) = 0$, so $H_2(S_j)$ is torsion-free; Euler characteristic zero gives its rank as two.

Transfer gives $\pi_{j*}\pi_j^* = m_j$; hence π_j^* is injective and its image lies in the T_j-invariants. The Lyndon–Hochschild–Serre spectral sequence for $1 \to \Lambda \to \Gamma_j \to \mathbb{Z}/m_j \to 1$ [20, Chapter VII, §6] has

$$E_2^{p,q} = H^p(\mathbb{Z}/m_j; \textstyle\bigwedge^q V).$$

It computes $H^*(S_j; \mathbb{Z})$ because the universal cover of this affine torus quotient is $\mathbb{R}^4$, hence contractible. Its transgression on an invariant $\xi \in V$ evaluates the extension class: $d_2\xi = \xi(v_j) \bmod m_j$. Since $V^{T_j} = \langle \gamma, \eta_j \rangle$ and $\eta_j(v_j) = 0$, the degree-one images and indices in (3.8) follow.

In degree two, the integral invariant lattices are

$$(\textstyle\bigwedge^2 V)^{T_j} = \langle \gamma\eta_j, \mathbf{q} \rangle$$

with no omitted finite-index overlattice. Indeed, writing an invariant as (a,b,c,d,e,f) in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$, the T_1 equations are $e=f=0$, $a=2b$, $c=3b+6d$, while the T_2 equations are $e=f=0$, $a=b$, $c=2b+6d$. Using $\gamma uw\delta$ as orientation,

$$(\gamma\eta_j)^2 = 0, \quad (\gamma\eta_1)\mathbf{q} = 3, \quad (\gamma\eta_2)\mathbf{q} = 2, \quad \mathbf{q}^2 = 12. \tag{3.10}$$

Let $L_j = \pi_j^* H^2(S_j)$. The intersection lattice of S_j is unimodular, and pullback multiplies its form by m_j. Hence $|\det L_j| = m_j^2$. The two invariant Gram determinants from (3.10) are 9 and 4. If k_j is the index of L_j in the invariant lattice, $k_j^2 |\det(\bigwedge^2 V)^{T_j}| = m_j^2$, so $k_1 = 1$ and $k_2 = 2$.

For $j=2$, multiplicativity of the spectral sequence gives $d_2(\gamma\eta_2) = -[\eta_2] \neq 0$ in $\langle \gamma, \eta_2\rangle / \langle 4\gamma, 2\eta_2\rangle$. Thus $\gamma\eta_2 \notin L_2$. The remaining index-two possibilities are $L_2 = \langle 2\gamma\eta_2, a\gamma\eta_2 + \mathbf{q}\rangle$, $a \in \{0,1\}$. The covering is étale, so $\pi_2^* K_{S_2}$ is trivial and $K_{S_2}^{\otimes 4}$ is trivial. Because $H^2(S_2;\mathbb{Z})$ is torsion-free, $c_1(S_2) = 0$ and $w_2(S_2) = 0$; the intersection form is even. Dividing the pullback pairings by four gives

$$\begin{pmatrix} 0 & 1 \\ 1 & a+3 \end{pmatrix},$$

so $a+3$ is even and $a = 1$. This proves (3.8).

For clarity, the displayed differential contains no suppressed coefficient calculation. For the cyclic group generated by T_2,

$$H^2(\mathbb{Z}/4; V) = V^{T_2}/(I + T_2 + T_2^2 + T_2^3)V = \langle \gamma, \eta_2\rangle / \langle 4\gamma, 2\eta_2\rangle.$$

The extension class gives $d_2\gamma = \gamma(v_2) = -1$ and $d_2\eta_2 = \eta_2(v_2) = 0$; the derivation rule therefore gives $d_2(\gamma\eta_2) = -[\eta_2]$, which is nonzero because $\eta_2 \notin \langle 4\gamma, 2\eta_2\rangle$.

In degree four the index is the covering degree. For degree three, direct exterior calculation gives the primitive invariant bases

$$(\textstyle\bigwedge^3 V)^{T_1} = \langle \gamma uw,\ uw\delta - 4\gamma u\delta - 2\gamma w\delta\rangle,$$
$$(\textstyle\bigwedge^3 V)^{T_2} = \langle \gamma uw,\ uw\delta - 3\gamma u\delta - 3\gamma w\delta\rangle.$$

Against (γ, η_j) their pairing matrices are

$$\begin{pmatrix} 0 & 1 \\ -3 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix},$$

so the absolute determinants are 3 and 2. If the degree-three index is r_j, the degree-one indices already proved are 3 and 4. Pullback multiplies the perfect $H^1(S_j) \times H^3(S_j)$ pairing by m_j, whose rank-two determinant is m_j^2. Hence $3 \cdot 3r_1 = 3^2$ and $2 \cdot 4r_2 = 4^2$; hence $r_1 = 1$, $r_2 = 2$. Together with degrees zero through two this proves (3.7). □

5.4 The terminal presentation and degree-two map

Proposition 5.5 (terminal group, including the nonabelian step)**.** ***[VERIFIED]*** *Consider*

$$G(\ell_0,\ell_1,\ell_2) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2}\rangle$$

and put $p = 12\ell_0 - 4\ell_1 - 3\ell_2$. *If* $p \neq 0$, *then* $G \cong \mathbb{Z}/|p|\mathbb{Z}$; *if* $p = 0$, *then* $G \cong \mathbb{Z}$. *For* $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$ *one has* $p = -1$ *and* $G = 1$.

Proof. The relation $xy = c^{\ell_0}$ gives $y = x^{-1}c^{\ell_0}$. Since c is central, x and y commute, so the apparently nonabelian presentation is already abelian. In the ordered additive generators (x, c) the remaining relations form

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -p.$$

The greatest common divisor of all entries of R is 1 because it divides both 3 and 4. Thus its Smith form is $\mathrm{diag}(1, |p|)$ when $p \neq 0$ and $\mathrm{diag}(1, 0)$ when $p = 0$. The stated groups follow. □

Proposition 5.6 (terminal degree-two lattice)**.** ***[VERIFIED]*** *Let $A : \mathbb{Z}^6 \to \mathbb{Z}^4$ have columns*

$$c_1 = (2,-4,-2,3)^t, \quad c_2 = (1,-2,-2,3)^t,$$
$$c_3 = (3,0,-4,0)^t, \quad c_4 = (0,1,0,-1)^t, \qquad c_5 = c_6 = 0.$$

Then $\operatorname{im} A$ *is saturated of rank three,*

$$\operatorname{im} A = \ker(4,2,3,2), \qquad \operatorname{coker} A \cong \mathbb{Z}, \tag{3.11}$$

and $(4,2,3,2)$ *is the primitive left-kernel generator.*

Proof. The row $\Phi = (4,2,3,2)$ annihilates the four nonzero columns:

$$\Phi c_1 = 8-8-6+6 = 0, \quad \Phi c_2 = 4-4-6+6 = 0, \quad \Phi c_3 = 12-12 = 0, \quad \Phi c_4 = 2-2 = 0.$$

It is primitive since the gcd of its entries is 1, so $\ker \Phi \subset \mathbb{Z}^4$ is a saturated rank-three lattice. The 3×3 minor on rows $(1,2,3)$ and columns $(1,2,4)$ is 2, whereas the minor on rows $(1,2,4)$ and the same columns is -3. Consequently the ideal generated by all maximal minors contains $(2,3) = (1)$. This proves that A has rank three and saturated image. We have $\operatorname{im} A \subseteq \ker \Phi$ between rank-three lattices. The finite quotient $\ker \Phi / \operatorname{im} A$ injects into the torsion-free group $\mathbb{Z}^4 / \operatorname{im} A$, so it is zero. Thus equality holds, and the primitive surjection $\Phi : \mathbb{Z}^4 \to \mathbb{Z}$ induces (3.11). □

Geometric boundary. Everything in this section is a theorem about the printed lattices, presentations, and matrices. It does not prove that the analytic gluing maps induce these representatives. That geometric identification is stated separately in theorems 10.4 and 10.5.

6 The analytic threefold constructed in the candidate

This section audits the construction itself. It does not infer existence from the later topology. The period family, the three local fillings, and the gluing are proved in that order. The notation is the candidate's: $B \simeq \mathbb{P}^1$, $B^\circ = B \setminus \{p_0, p_1, p_2\}$, and the local monodromy orders at p_1, p_2 are $m_1 = 3, m_2 = 4$.

6.1 The period family

Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ be the orientation-preserving $(3,4,\infty)$-triangle group acting on a simply connected Riemann surface $\mathfrak{h}_z$, with elliptic generators g_1, g_2 and $g_0 = (g_1 g_2)^{-1}$. Let t be the affine coordinate on B with values $0, 1, \infty$ at p_1, p_2, p_0, respectively. Write

$$\pi : \mathfrak{h}_z \longrightarrow B \setminus \{p_0\}$$

for the orbifold quotient map; its stabilizers at chosen lifts $z_j \in \pi^{-1}(p_j)$ are $\langle g_j \rangle$. Let $\rho_V : \Delta \to \mathrm{GL}(V)$ be the printed representation, $\rho_V(g_j) = T_j$, and put

$$M_g := \rho_V(g)^{\mathsf{t}}, \qquad A_g := M_g^{-1}.$$

Thus $M_{gh} = M_h M_g$, $A_{g_j} = A_j$, and $A_{g_0} = M_0$ in the notation of theorem 5.1.

Proposition 6.1 (existence and non-degeneracy of the period matrix)**.** ***[VERIFIED]*** *There are holomorphic functions*

$$\tau : \mathfrak{h}_z \to \mathfrak{h}, \qquad \mu, \beta : \mathfrak{h}_z \to \mathbb{C}$$

satisfying

$$\begin{aligned} &\tau \circ g_0 = \tau - 1, \quad \mu \circ g_0 = \mu, && \beta \circ g_0 = \beta + 1, \\ &\tau \circ g_2 = -1/\tau, \quad \mu \circ g_2 = 1 + \mu/\tau, && \beta \circ g_2 = \beta - 3 - 6\mu^2/\tau, \end{aligned} \tag{2.1}$$

and the corresponding order-three transformation laws obtained from $\tau \circ g_1 = (\tau - 1)/\tau$. They may be chosen so that

$$\operatorname{Im}\tau > 0, \qquad D := \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} < 0. \tag{2.2}$$

Consequently

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix} \tag{2.3}$$

maps the marked rank-four lattice Λ isomorphically onto a lattice in $\mathbb{C}^2$ at every point. The quotients $\mathbb{C}^2/\Pi(z)\Lambda$ form a holomorphic family of compact two-dimensional complex tori over $\mathfrak{h}_z$, equivariant for Δ.

Proof. Normalize the modular function by $j(q) = q^{-1}(1 + O(q))$ and put $J_B = 1728(t \circ \pi)$; the modular facts used here are collected in [19, Chapter VII]. Away from the inverse images of 0 and 1728, the classical modular map is an unramified covering with deck group $\mathrm{PSL}_2(\mathbb{Z})$. A meridian at p_1 maps under J_B to the third power of a meridian whose modular monodromy has order three. At p_2 it maps to the fourth power of one whose modular monodromy has order two. Thus every normal generator of the punctured fundamental group has trivial deck image, and the lifting criterion gives τ with $j(\tau) = J_B$. Composing with a disc coordinate on the upper half-plane gives a bounded holomorphic map near each deleted point; the removable-singularity theorem and the maximum principle extend the lift through those points.

Normalize $\tau(z_1) = \rho = e^{\pi i/3}$. Local ramification gives a simple zero of $\tau - \rho$ at z_1 and a double zero of $\tau - i$ at z_2. Hence the order-three generator acts by $S_1(\tau) = (\tau - 1)/\tau$. The order-four generator cannot act trivially: otherwise the invariant local power series $\tau - i$ would have order divisible by four, contrary to its order two. Its modular image therefore has order two.

It remains to determine the cusp image. It is nontrivial: if τ were single-valued on the punctured cusp disc, then $q = e^{2\pi i\tau}$, which satisfies $0 < |q| < 1$, would extend across the puncture. The expansion $j = q^{-1}(1 + O(q))$ and $|j| \to \infty$ force the extended value to be $q(0) = 0$; the alternative $q(0) \neq 0$ would keep j bounded. But q would simultaneously possess the single-valued logarithm $2\pi i\tau$, contradicting the positive winding number of a zero. For sufficiently large $|j|$, each connected component of its inverse image lies between two horoballs based at one cusp. The stabilizer of that component fixes the cusp, so the nontrivial cusp image is parabolic. Write an integral lift as

$$P = I + k \begin{pmatrix} -pr & p^2 \\ -r^2 & pr \end{pmatrix}, \qquad \gcd(p, r) = 1.$$

The condition that $S_1^{-1}P^{-1}$ have order two is $-k(p^2 - pr + r^2) = 1$. Positive definiteness of the quadratic form forces $k = -1$ and $p^2 - pr + r^2 = 1$. Conjugating by a power of S_1 gives $P(\tau) = \tau - 1$, and then the other generator is $S(\tau) = -1/\tau$. At the cusp, $q = t_c u_1(t_c)$ with $u_1(0) \neq 0$, because the pole of j is simple.

We now solve the two affine cocycle problems, including the orbifold and cusp regularity which the abbreviated argument had suppressed. Define $\widetilde{\jmath} : \Delta \times \mathfrak{h}_z \to \mathbb{C}^*$ on the generators by

$$\widetilde{\jmath}(g_1, z) = -\tau(z), \qquad \widetilde{\jmath}(g_2, z) = \tau(z),$$

and extend it by $\widetilde{\jmath}(gh, z) = \widetilde{\jmath}(g, hz)\widetilde{\jmath}(h, z)$. This is well defined: the products around the two finite cyclic subgroups are

$$(-1)^3\tau\frac{\tau - 1}{\tau}\frac{-1}{\tau - 1} = 1, \qquad \tau\left(-\frac{1}{\tau}\right)\tau\left(-\frac{1}{\tau}\right) = 1.$$

Moreover $\widetilde{\jmath}(g_0, \cdot) = 1$.

For an open $U \subset B$, set $\widetilde{U} = \pi^{-1}(U \setminus \{p_0\})$. Define

$$\mathcal{L}(U) = \left\{\nu \in \mathcal{O}(\widetilde{U}) : \nu \circ g = \nu/\widetilde{\jmath}(g, \cdot) \quad (g \in \Delta)\right\}$$

and let $\mathcal{M}(U)$ be the holomorphic functions on $\widetilde{U}$ satisfying

$$\mu \circ g_1 = \frac{1-\mu}{\tau}, \qquad \mu \circ g_2 = 1 + \frac{\mu}{\tau}. \tag{2.1a}$$

If $p_0 \in U$, both definitions additionally require that the restriction to the distinguished $\langle g_0 \rangle$-invariant cusp component, viewed as a function of t_c, extend holomorphically across $t_c = 0$. Because $\widetilde{j}(g_0) = 1$ and $\mu \circ g_0 = \mu$, this requirement is meaningful and is stable under restriction. Thus $\mathcal{L}$ is a line sheaf and $\mathcal{M}$, if locally nonempty, is an affine $\mathcal{L}$-torsor.

Local nonemptiness is explicit. Over an ordinary small disc, prescribe $\mu = 0$ on one sheet and extend using (2.1a). At the cusp prescribe $\mu = 0$ on the distinguished component. At the elliptic points use

$$\mu_1 = \frac{2-\tau}{3} \quad \text{on } \Delta_1, \qquad \mu_2 = \frac{1-\tau}{2} \quad \text{on } \Delta_2. \tag{2.1b}$$

Indeed

$$\mu_1 \circ g_1 = \frac{\tau+1}{3\tau} = \frac{1-\mu_1}{\tau}, \qquad \mu_2 \circ g_2 = \frac{\tau+1}{2\tau} = 1 + \frac{\mu_2}{\tau}.$$

The cyclic products above, or direct iteration of (2.1a), show that these prescriptions close after three and four steps.

To identify $\mathcal{L}$, use that $E_6 \circ \tau$ has zeros of even order on the simply connected surface $\mathfrak{h}_z$, and fix a holomorphic square root. Put

$$\Theta = \left(\frac{E_4^2 E_6^{1/2}}{\Delta_{\mathrm{mod}}} \right) \circ \tau.$$

The modular weights $8, 3, 12$, with the square-root sign checked in the linearizing coordinate at z_2, give the automorphy law of $\mathcal{L}$. The orders of Θ at z_1, z_2 are two and one, and $\Theta = t_c^{-1} \cdot (\text{unit})$ at the distinguished cusp. Taking the stabilizer orders into account, division by Θ identifies the ordinary and elliptic stalks with $\mathcal{O}_B$, while the cusp stalk is $\mathfrak{m}_{p_0}\Theta$. Hence

$$\mathcal{L} \simeq \mathcal{O}_B(-p_0) simeq \mathcal{O}_{\mathbb{P}^1}(-1).$$

Consequently $H^1(B, \mathcal{L}) = H^0(B, \mathcal{L}) = 0$: the locally nonempty torsor $\mathcal{M}$ has a unique global section. Evaluating at the fixed points gives

$$\mu(z_1) = \frac{2-\rho}{3}, \qquad \mu(z_2) = \frac{1-i}{2},$$

and the cusp condition says precisely that its invariant branch extends holomorphically.

For the second affine problem put

$$\phi_1 = 2 - \frac{6(1-\mu)^2}{\tau}, qquad \phi_2 = -3 - \frac{6\mu^2}{\tau}.$$

Substitution around the finite orbits gives

$$\sum_{k=0}^{2} \phi_1 \circ g_1^k = 0, \qquad \sum_{k=0}^{3} \phi_2 \circ g_2^k = 0. \tag{2.1c}$$

For example, the three summands in the first identity are

$$2 - \frac{6(1-\mu)^2}{\tau}, \quad 2 - \frac{6(\tau-1+\mu)^2}{\tau(\tau-1)}, \quad 2 + \frac{6\mu^2}{\tau-1},$$

and their numerator is $6\tau(\tau-1)$; for the order-four identity the bracket after extracting $6/\tau$ is 2τ, cancelling -12.

Define $\mathcal{B}(U)$ to consist of holomorphic β on $\widetilde{U}$ with $\beta \circ g_j - \beta = \phi_j$; at the cusp require $b = \beta + \tau$ to extend holomorphically in t_c. Differences of sections are ordinary holomorphic functions on U, so $\mathcal{B}$ is a pseudo-torsor under $\mathcal{O}_B$. It is locally nonempty: prescribe $\beta = 0$ on one ordinary sheet; prescribe $\beta = -\tau$ on the distinguished cusp component; and at p_j use

$$\beta_j = \frac{1}{m_j} \sum_{k=0}^{m_j-1} k\,(\phi_j \circ g_j^k). \tag{2.1d}$$

Equation (2.1c) makes the last expression telescope to $\beta_j \circ g_j - \beta_j = \phi_j$. Therefore $\mathcal{B}$ is an $\mathcal{O}_B$-torsor. Since $H^1(B, \mathcal{O}_B) = 0$, it has global sections, and any two differ by a constant; by construction $b = \beta + \tau$ is invariant and holomorphic at the cusp.

The real determinant is

$$\det_{\mathbb{R}} \Pi = -\det \operatorname{Im} Z = \operatorname{Im} \tau\, D. \tag{2.4}$$

For completeness, the equivariance can be checked before any quotient is taken. Regard the rows of Π as the two covectors

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta \quad \text{in } V \otimes \mathbb{C}. \tag{2.3a}$$

Multiplication by the printed matrices gives

$$\begin{aligned} T_1\sigma_1 &= (6\mu - 6)\gamma + (1-\tau)u - \tau w, \\ T_1\sigma_2 &= (\beta + 2)\gamma + (1-\mu)u + (1-\mu)w + \delta, \\ T_2\sigma_1 &= (6\mu + 6\tau)\gamma - u + \tau w, \\ T_2\sigma_2 &= (\beta + 6\mu - 3)\gamma + u + \mu w + \delta. \end{aligned} \tag{2.3b}$$

Using the four affine transformation laws in (2.1a)–(2.1d), coefficient comparison in (2.3b) yields

$$\begin{aligned} \sigma_1 \circ g_1 &= -\tau^{-1} T_1\sigma_1, \quad \sigma_2 \circ g_1 = T_1\sigma_2 + (1-\mu)\tau^{-1} T_1\sigma_1, \\ \sigma_1 \circ g_2 &= \tau^{-1} T_2\sigma_1, \qquad \sigma_2 \circ g_2 = T_2\sigma_2 - \mu\tau^{-1} T_2\sigma_1. \end{aligned} \tag{2.3c}$$

For instance, dividing the first line of (2.3b) by $-\tau$ gives the coefficients $6(1-\mu)/\tau$, $(\tau - 1)/\tau$, and 1, exactly those of $\sigma_1(g_1 z)$; subtracting $(1-\mu)$ times that normalized row from $T_1\sigma_2$ gives $\sigma_2(g_1 z)$. The two g_2 identities follow by the same coefficient check. Thus, with

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix},$$

pairing (2.3c) with $\lambda \in \Lambda$ gives

$$\Pi(gz) = R_g(z)\Pi(z)M_g. \tag{2.3d}$$

Since $M_{gh} = M_h M_g$, uniqueness of the left factor gives $R_{gh}(z) = R_g(hz)R_h(z)$; hence these formulae define an action, not merely generator-wise maps. Taking real determinants in (2.3d), using $\det M_g = 1$ and $\operatorname{Im}\tau(g_j z) = \operatorname{Im}\tau/|\tau|^2$, shows $D(gz) = D(z)$. At the cusp,

$$D = \operatorname{Im} b - \operatorname{Im}\tau - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \longrightarrow -\infty.$$

The descended function is bounded above on the complement of a cusp neighbourhood. Replacing β by $\beta + c_0$ adds $\operatorname{Im} c_0$ to D, so a sufficiently negative imaginary constant makes (2.2) hold globally. Equation (2.4) then makes Π a real isomorphism everywhere. The equivariance identity supplies the claimed Δ-action and completes the construction of the period family. □

For the cusp filling, record the local periods explicitly. Let t_c be the coordinate vanishing at p_0. The simple pole of j gives

$$q = e^{2\pi i\tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

After choosing a logarithm of u_1, put

$$h = (2\pi i)^{-1} \log u_1, \qquad s = \tau - h, \qquad b = \beta + \tau.$$

Then $e^{2\pi i s} = t_c$, $s \circ g_0 = s - 1$, and μ, b, h are holomorphic functions of t_c at the origin. Consequently

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z = sB_0 + C(t_c), \qquad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}, \tag{2.5a}$$

and C is holomorphic on a full cusp disc. In particular, for $\bar{\lambda} \in \bar{\Lambda} \simeq \mathbb{Z}^2$,

$$\exp(2\pi i Z\bar{\lambda}) = t_c^{B_0\bar{\lambda}} c_{\bar{\lambda}}(t_c), \qquad c_{\bar{\lambda}}(t_c) = \exp(2\pi i C(t_c)\bar{\lambda}). \tag{2.5b}$$

6.2 The toric filling at the cusp

Let $\mathcal{T}_{A_2}$ be the translation-invariant triangulation of $\mathbb{R}^2$ by the two triangles $\{v, v + e_1, v + e_2\}$ and $\{v + e_1, v + e_2, v + e_1 + e_2\}$. In $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$, take the cones over all cells at height one, and denote the associated toric threefold by Y. We use the standard orbit–cone and affine-chart constructions of [16, Chapters 1 and 3]; all nonstandard assertions caused by the infinite fan are proved directly below.

For $\bar{\lambda} \in \bar{\Lambda}$, define the integral linear map

$$\varphi_{\bar{\lambda}}(y, y_3) = (y + y_3 B_0\bar{\lambda}, y_3).$$

On the height-one slice this is translation by $B_0\bar{\lambda} \in \mathbb{Z}^2$, so it preserves the triangulation and maps the cone over a cell P to the cone over $P + B_0\bar{\lambda}$. It therefore induces a toric automorphism $\Phi_{\bar{\lambda}}$ of Y, satisfying

$$\Phi_{\bar{\lambda}}(x, t) = (t^{B_0\bar{\lambda}} x, t)$$

on the dense torus. Because t and $C(t)$ are holomorphic on the full cusp model, multiplication by $(c_{\bar{\lambda}}(t), 1) \in (\mathbb{C}^*)^3$ defines a biholomorphism on all of Y_ϵ. Set

$$\Psi_{\bar{\lambda}}(p) = (c_{\bar{\lambda}}(t(p)), 1) \cdot \Phi_{\bar{\lambda}}(p).$$

The maps $\Phi_{\bar{\lambda}}$ fix the first two cocharacter directions and preserve t, so they commute with this torus multiplication. The linearity of $C(t)\bar{\lambda}$ gives $c_{\bar{\lambda}} c_{\bar{\mu}} = c_{\bar{\lambda}+\bar{\mu}}$; hence $\Psi_{\bar{\lambda}}\Psi_{\bar{\mu}} = \Psi_{\bar{\lambda}+\bar{\mu}}$. This proves, before taking any quotient, both the holomorphic extension and the action law used below.

For a triangle $\sigma = \{v_0, v_1, v_2\}$, let θ_i^σ be its barycentric coordinates and let $z_i = \chi^{m_i}$ be the coordinates dual to $(v_i, 1)$. Set

$$\sigma^{(\eta)} = \{y : \theta_i^\sigma(y) > -\eta \text{ for } i = 0, 1, 2\}, \quad \mathcal{U}_\sigma(S) = \{|z_i| < S \text{ for all } i\}.$$

Lemma 6.2 (toric chart control)**.** *For a torus point $p = (x, t)$ with $0 < |t| < 1$, put*

$$y(p) = \frac{(\log|x_1|, \log|x_2|)}{\log|t|}.$$

Then

$$p \in \mathcal{U}_\sigma(S) \quad \Longleftrightarrow \quad y(p) \in \sigma^{(\log S/|\log|t||)}. \tag{2.5c}$$

Every point of $\{|t| < 1\} \subset Y$ belongs to $\overline{\mathcal{U}}_\sigma(1) = \{|z_i| \leq 1\}$ for some σ. Consequently the sets $\mathcal{U}_\sigma(2)$ cover Y_ϵ, and every compact subset is contained in the union of finitely many members of this cover.

Proof. Since $(y,1) = \sum_i \theta_i^\sigma(y)(v_i,1)$, duality gives

$$\log|z_i(p)| = \log|t(p)|\,\theta_i^\sigma(y(p)).$$

Because $\log|t| < 0$, this is exactly (2.5c).

For the boundary assertion let p lie in the toric orbit $O_{\widehat{P}}$ of the cone over a cell P. In a chart with $\sigma \supseteq P$, the coordinates corresponding to vertices of P vanish. On the quotient lattice $N'/(N' \cap \operatorname{span}\widehat{P})$, the remaining coordinates are the characters dual to the image cone $\bar\sigma$. Hence $p \in \overline{\mathcal{U}}_\sigma(1)$ exactly when $-\operatorname{Log}_{\widehat{P}}(p) \in \bar\sigma$. The cones $\bar\sigma$, for triangles $\sigma \supseteq P$, cover the quotient real vector space: lift a vector n, choose a relative interior point $p_0 \in P \times \{1\}$, and for sufficiently small $u > 0$ place the height-one rescaling of $p_0 + un$ in a triangle of the locally finite triangulation. That triangle contains P, and its quotient cone contains n. This proves the asserted cover. Compactness then supplies a finite subcover. □

Theorem 6.3 (the cusp quotient and its central fibre)**.** ***[VERIFIED]*** *For the period functions and holomorphic cusp matrix $C(t)$ constructed above, and sufficiently small $\epsilon > 0$, the maps*

$$\Psi_{\bar\lambda}(x,t) = (c_{\bar\lambda}(t)t^{B_0\bar\lambda}x, t), \qquad c_{\bar\lambda}(t) = \exp(2\pi i C(t)\bar\lambda), \tag{2.5}$$

extend to a free properly discontinuous action of $\bar\Lambda \simeq \mathbb{Z}^2$ on $Y_\epsilon = \{|t| < \epsilon\}$. The quotient

$$f_0 : N_0 = Y_\epsilon/\bar\Lambda \longrightarrow \Delta_\epsilon$$

is a proper holomorphic map from a connected smooth complex threefold. Its nonzero fibres are the tori of theorem 6.1. Its full scheme-theoretic central fibre W is reduced and has local equations

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2. \tag{2.6}$$

The surface W is irreducible and nonnormal. Its normalization is $\nu : E_0 \simeq \mathrm{dP}_6 \to W$, obtained by identifying opposite sides of the toric hexagon; its double locus consists of three rational curves through exactly two triple points. Thus W is a normal-crossings surface, but it is not a simple-normal-crossings divisor.

Proof. For a bounded cell P, the cone over $P \times \{1\}$ is strongly convex, and two such cones meet in the cone over the common face of the two cells. The locally finite triangulation therefore defines a countable rational fan with support $(\mathbb{R}^2 \times \mathbb{R}_{>0}) \cup \{0\}$. For a triangle with vertices v_0, v_1, v_2, the determinant of its three height-one generators is $+1$ for the first kind and -1 for the second. Every cone is therefore unimodular. The ordinary pairwise affine-chart separation proof for a toric fan applies without a finiteness hypothesis; countability of the fan gives second countability. Thus Y is a connected Hausdorff smooth complex threefold with dense torus.

If m_0, m_1, m_2 is the basis dual to the height-one generators, then the corresponding affine chart is $\mathbb{C}^3$, with coordinates $z_i = \chi^{m_i}$, and

$$t = z_0z_1z_2. \tag{2.7}$$

Since every primitive ray generator has height one, the divisor of t is $\sum_v E_v$, all with multiplicity one. Equation (2.7) proves that the scheme-theoretic fibre $t^{-1}(0)$ is reduced and normal crossings. The star of a height-one vertex has rays $e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2$, the complete smooth hexagonal fan; hence every irreducible component E_v is dP_6.

Write $R(t) = -2\pi \operatorname{Im} C(t)$, which is bounded because C is holomorphic at zero. Choose ϵ such that $|\log\epsilon| > 2\|R\|_\infty$. At a torus point fixed by a nonzero $\bar\lambda$, taking logarithmic absolute values in (2.5) gives

$$\log|t|B_0\bar\lambda + R(t)\bar\lambda = 0,$$

impossible because $|B_0\bar\lambda| = |\bar\lambda|$ and the first term has more than twice the norm of the second. On $t = 0$, a cell is translated by $B_0\bar\lambda \neq 0$; a bounded cell cannot be fixed by a nonzero translation. Thus the action is free.

For proper discontinuity put

$$B_t = B_0 + \frac{R(t)}{\log|t|}.$$

Then $\|B_t - B_0\| \le 1/2$, so $|B_t\bar\lambda| \ge |\bar\lambda|/2$. In logarithmic position coordinates, (2.5) is

$$y(\Psi_{\bar\lambda}p) = y(p) + B_t\bar\lambda. \tag{2.8}$$

Put $\eta_0 = \log 2/|\log\epsilon|$. If $\Psi_{\bar\lambda}\mathcal{U}_\sigma(2)$ meets $\mathcal{U}_{\sigma'}(2)$, the intersection is open and hence contains a point of the dense invariant torus. By theorem 6.2, for that point p and its image p',

$$y(p) \in \sigma^{(\eta_0)}, \qquad y(p') \in \sigma'^{(\eta_0)}.$$

Both thickenings are bounded, so (2.8) gives

$$\frac{1}{2}|\bar\lambda| \le |B_t\bar\lambda| = |y(p') - y(p)| \le \sup_{a\in\sigma'^{(\eta_0)},\, b\in\sigma^{(\eta_0)}} |a-b|.$$

Thus only finitely many $\bar\lambda$ can make this fixed pair of charts meet. A compact set is contained in finitely many of the charts from theorem 6.2; taking the finite union over all chart pairs proves proper discontinuity. Freeness and proper discontinuity give a Hausdorff complex-manifold quotient and a holomorphic covering map.

To prove properness, fix $\epsilon_1 < \epsilon$. Let $\mathcal{F}_0$ be the finite set of triangles meeting the closed Euclidean ball of radius $\sqrt{2}$, and put

$$K_0 = \bigcup_{\sigma\in\mathcal{F}_0} \left(\overline{\mathcal{U}}_\sigma(1) \cap \{|t| \le \epsilon_1\}\right).$$

This is compact. For a torus point p with $0 < |t(p)| \le \epsilon_1$, put $u = B_{t(p)}^{-1}y(p)$ and choose $\bar\lambda \in \mathbb{Z}^2$ with $|u + \bar\lambda|_\infty \le 1/2$. Since $\|B_t\| \le 3/2$, equation (2.8) gives

$$|y(\Psi_{\bar\lambda}p)| = |B_t(u+\bar\lambda)| \le \frac{3}{2\sqrt{2}} < \sqrt{2}.$$

Choose a triangle containing this point. Formula (2.5c) with $S = 1$ places the translate in the corresponding $\overline{\mathcal{U}}_\sigma(1) \subset K_0$.

If p lies in the central orbit belonging to a cell P, translate a vertex of P to zero; this is possible because $B_0\bar\Lambda = \mathbb{Z}^2$. The refined boundary part of theorem 6.2 then places the translate in $\overline{\mathcal{U}}_\sigma(1)$ for a triangle σ containing zero, hence in K_0. Therefore the image of K_0 in the quotient is exactly $f_0^{-1}(\{|t| \le \epsilon_1\})$, which is compact. Every compact subset of the base lies in such a closed subdisc, so f_0 is proper.

For $t \neq 0$, choose s with $e^{2\pi i s} = t$. The multiplier in (2.5) is $\exp(2\pi i Z(s)\bar\lambda)$, so exponentiation identifies the fibre with

$$\mathbb{C}^2/(\mathbb{Z}^2 + Z(s)\mathbb{Z}^2).$$

It is compact because $\operatorname{Im} Z(s)$ is invertible by (2.2).

The quotient map is a local biholomorphism; therefore the local equation (2.7), its multiplicity one, and reducedness descend to W. Every cell has a vertex, and $B_0\bar\Lambda = \mathbb{Z}^2$ acts simply transitively on vertices, so every orbit in the central fibre meets E_0. Thus $E_0 \to W$ is surjective. Two interior points cannot be identified; on the boundary, precisely opposite edges and the corresponding hexagon vertices are identified. The fibres have at most three points. Since E_0 is compact, the map is finite; since it is generically injective from a smooth irreducible surface to the reduced surface W, it is the normalization. The two translation-orbits of triangles give the two triple points, and the three unoriented edge directions give the three double curves. Irreducibility follows from the finite surjection by E_0. □

Proposition 6.4 (the cusp punctured-overlap biholomorphism)**.** ***[VERIFIED]*** *Let $D_0^* = \{0 < |t_c| < \epsilon\}$, and shrink it if necessary so that its inverse image has a distinguished component U_0 stabilized exactly by $\langle g_0 \rangle$. Exponentiation induces a fibre-preserving biholomorphism*

$$J|_{D_0^*} \xrightarrow{\sim} N_0|_{D_0^*} = N_0 \setminus W. \tag{2.8i}$$

On the covering space it is given by

$$\mathcal{E} : U_0 \times \mathbb{C}^2 \longrightarrow (\mathbb{C}^*)^2 \times D_0^*, \qquad \mathcal{E}(z, \zeta_1, \zeta_2) = \big(e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}\big). \tag{2.8ii}$$

Proof. First identify every deck transformation. From $s \circ g_0 = s - 1$, the equality $t_c \circ \pi = e^{2\pi i s}$, and the fact that t_c is a base coordinate, ds never vanishes on U_0. Thus s is a local biholomorphism to the logarithmic cover of D_0^*. The restriction $U_0 \to D_0^*$ is a connected regular cover with deck group $\langle g_0 \rangle \simeq \mathbb{Z}$. Under the classification of connected covers of the punctured disc, it corresponds to a subgroup of $\pi_1(D_0^*) \simeq \mathbb{Z}$; because its deck group is infinite, that subgroup is zero. Hence U_0 is the universal cover. The logarithmic cover is universal as well, and its deck generator acts by $s \mapsto s - 1$. Therefore s is an equivariant biholomorphism onto that logarithmic cover.

The cusp transformation laws give

$$\Pi(g_0 z) = [Z(s-1) \mid I_2] = [Z(s) - B_0 \mid I_2].$$

In the same block decomposition, the explicit matrix in theorem 5.1 gives

$$M_{g_0} = M_0^{-1} = \begin{pmatrix} I_2 & 0 \\ -B_0 & I_2 \end{pmatrix}.$$

Consequently

$$\Pi(g_0 z) = \Pi(z) M_{g_0}.$$

Comparison with (2.3d) and real invertibility of $\Pi(g_0 z)$ yields $R_{g_0} = I_2$. Hence the lifted cusp generator acts by $(z, \zeta) \mapsto (g_0 z, \zeta)$, and $\mathcal{E}$ is invariant under it.

Let $\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$. Translation by this sublattice adds an integral vector to ζ, so it also lies in the deck group of $\mathcal{E}$. Conversely, if two points have the same image under $\mathcal{E}$, their s-coordinates differ by an integer and their ζ-coordinates by an element of $\mathbb{Z}^2$. Equivariance and injectivity of the logarithmic coordinate then show that they differ by a unique element of

$$H = \Lambda_{\mathrm{tor}} \times \langle g_0 \rangle.$$

Thus $\mathcal{E}$ identifies $(U_0 \times \mathbb{C}^2)/H$ with $(\mathbb{C}^*)^2 \times D_0^*$.

It remains to identify the residual action. Write $\bar{\Lambda} = \Lambda / \Lambda_{\mathrm{tor}}$, and let λ be any lift of $\bar{\lambda}$. In the period basis, translation by λ adds $Z(s)\bar{\lambda}$ to ζ, up to an integral vector killed by exponentiation. Therefore (2.5b) gives

$$\mathcal{E}(z, \zeta + \Pi(z)\lambda) = \big(c_{\bar{\lambda}}(t_c) t_c^{B_0 \bar{\lambda}} x, t_c\big) = \Psi_{\bar{\lambda}}(\mathcal{E}(z, \zeta)). \tag{2.8iii}$$

This is independent of the chosen lift. Moreover M_0 is the identity on $\bar{\Lambda}$, since $(M_0 - I)\Lambda \subset \Lambda_{\mathrm{tor}}$, so the quotient of the full cusp deck group by H is precisely the additive lattice $\bar{\Lambda}$. Dividing (2.8iii) by that residual action gives

$$(U_0 \times \mathbb{C}^2)/(\Lambda \rtimes \langle g_0 \rangle) \simeq ((\mathbb{C}^*)^2 \times D_0^*)/\bar{\Lambda}.$$

The left side is $J|_{D_0^*}$, while the right side is $N_0|_{D_0^*}$ because $t^{-1}(\mathbb{C}^*)$ is the dense torus of Y. This proves (2.8i), including holomorphicity of the inverse. □

6.3 The finite logarithmic fillings and the global gluing

Set

$$\varepsilon = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad \varepsilon' = \widehat{\gamma} + 3\widehat{u} - 3\widehat{w}. \tag{2.8a}$$

Direct multiplication by the matrices in theorem 5.1 gives $A_1\varepsilon = \varepsilon$ and $A_2\varepsilon' = \varepsilon'$.

A *linearizing pair* (Δ_j, s_j) consists of a simply connected g_j-invariant neighbourhood of z_j, disjoint from all its translates outside $\langle g_j\rangle$, and an injective coordinate satisfying

$$s_j(z_j) = 0, \qquad s_j \circ g_j = \zeta_j s_j, \qquad \zeta_j = e^{-2\pi i/m_j}. \tag{2.8b}$$

Shrinking gives such a pair. Put $D_j = \pi(\Delta_j)$, $\Delta_j^* = \Delta_j \setminus \{z_j\}$, and choose the base coordinate t_j by $t_j \circ \pi = s_j^{m_j}$. Let

$$\mathcal{T} = (\mathfrak{h}_z \times \mathbb{C}^2)/\Lambda, \qquad \mathfrak{h}_z^\circ = \mathfrak{h}_z \setminus (\Delta z_1 \cup \Delta z_2), \qquad J = (\mathcal{T}|_{\mathfrak{h}_z^\circ})/\Delta$$

$$J_j' = \mathcal{T}|_{\Delta_j}, \qquad A_0^{(j)} = \mathbb{C}^2/\Pi(z_j)\Lambda. \tag{2.8c}$$

For $b \in \Lambda \otimes \mathbb{R}$, define the varying holomorphic lift

$$\widetilde{g}_j^b(z,\zeta) = (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z)b). \tag{2.8d}$$

Equation (2.3d) says

$$R_{g_j}(z)\Pi(z) = \Pi(g_j z)A_j, \tag{2.8e}$$

so (2.8d) normalizes the lattice action and descends to J_j'. In the flat real-analytic coordinates

$$\Xi(z, x+\Lambda) = [z, \Pi(z)x]$$

one has, with $\Sigma_k = I + A_j + \cdots + A_j^{k-1}$,

$$\Xi^{-1}(\widetilde{g}_j^b)^k \Xi(z,x) = (g_j^k z, A_j^k x + \Sigma_k b). \tag{2.8f}$$

We call $\widetilde{g}_j^0$ the linear lift and, for $v \in \Lambda^{A_j}$, call $\widetilde{g}_j^{\log} := \widetilde{g}_j^{v/m_j}$ the logarithmic lift.

Proposition 6.5 (finite fillings)**.** ***[VERIFIED]*** *Let A_j be the two finite monodromies and choose*

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad \gamma(v_1) = 1, \quad \gamma(v_2) = -1.$$

On the local torus family over a linearizing disc Δ_j, the affine map

$$(z,x) \longmapsto (g_j z, A_j x + v_j/m_j) \tag{2.9}$$

generates a free action of order m_j. The quotient $f_j : N_j \to D_j$ is a smooth complex threefold with

$$f_j^*(p_j) = m_j S_j, \qquad S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j\rangle. \tag{2.10}$$

Over the punctured disc it is biholomorphic to the original period family.

Proof. First prove freeness on the actual local family. For a finite-order integral map A, the affine self-map $x \mapsto Ax + b$ of $(\Lambda \otimes \mathbb{R})/\Lambda$ has a fixed point exactly when

$$\varphi(b) \in \mathbb{Z} \quad \text{for every } \varphi \in \mathrm{Hom}(\Lambda, \mathbb{Z}) \text{ with } \varphi \circ A = \varphi. \tag{2.9a}$$

Necessity is immediate. Conversely, the averaging projector $p = m^{-1}\sum_{r=0}^{m-1} A^r$ has image $\ker(A-I)_{\mathbb{R}}$ and kernel $\mathrm{im}(A-I)_{\mathbb{R}}$; the dual lattice of $p(\Lambda)$ is exactly the invariant integral covector lattice. Thus (2.9a) puts pb in $p(\Lambda)$, and $b \in (A-I)(\Lambda \otimes \mathbb{R}) + \Lambda$, which supplies a fixed point.

The exact matrix calculation gives

$$V^{T_1^k} = \langle \gamma, \psi_1 \rangle, \quad \psi_1 = 2u + w + 3\delta \quad (k = 1, 2),$$

and

$$V^{T_2^k} = \langle \gamma, \psi_2 \rangle, \quad \psi_2 = u + w + 2\delta \quad (k = 1, 2, 3). \tag{2.9b}$$

For $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$,

$$\gamma(v_1) = 1, \quad \psi_1(v_1) = 0, \qquad \gamma(v_2) = -1, \quad \psi_2(v_2) = 0. \tag{2.9c}$$

Because $A_j v_j = v_j$, formula (2.8f) shows that the k-th power of the logarithmic lift has translation kv_j/m_j. For $j = 1$, (2.9a) fails for $k = 1, 2$ already on γ, since $k/3 \notin \mathbb{Z}$. For $j = 2$, it fails for $k = 1, 2, 3$, since $-k/4 \notin \mathbb{Z}$. Away from z_j, the base map g_j^k has no fixed point in Δ_j. Hence every nonidentity power is fixed-point free on J_j'. At $k = m_j$, (2.8f) is translation by $v_j \in \Lambda$, hence the identity; the order is exactly m_j because the map covers g_j.

It follows that

$$N_j = J_j'/\langle \widetilde{g}_j^{\log} \rangle$$

is a smooth complex threefold and the quotient map is finite étale of degree m_j. The function $s_j^{m_j} = t_j \circ \pi$ is invariant and descends to $f_j : N_j \to D_j$. The projection $J_j' \to \Delta_j$ is proper: over a compact subset, the image of the compact set obtained by letting x range over a fixed unit parallelotope in $\Lambda \otimes \mathbb{R}$ is the whole inverse image. Together with finiteness of $\Delta_j \to D_j$ and of the quotient, this proves properness of f_j. Pulling the divisor of t_j back through the etale quotient gives the divisor of $s_j^{m_j}$, namely m_j times the reduced central torus. This proves (2.10).

It remains to identify the punctured filling with the original family and to check that no logarithm branch has entered the answer. On a simply connected $U \subset \Delta_j^*$, choose a branch ℓ of $\log s_j$ and put

$$\sigma_j^\ell(z) = \frac{\ell(z)}{2\pi i} \Pi(z) v_j. \tag{2.9d}$$

Changing ℓ by $2\pi i n$ changes (2.9d) by the lattice vector $n\Pi(z)v_j$. The classes therefore glue to a branch-independent holomorphic section σ_j of $J_j'|_{\Delta_j^*}$; translation by that section defines a biholomorphism h_j.

For the conjugacy, use ℓ on U and on $g_j U$ use the branch

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}.$$

This is a logarithm of $s_j(w)$ because $s_j \circ g_j = e^{-2\pi i/m_j} s_j$. For $z \in U$, equation (2.8e) and $A_j v_j = v_j$ give

$$\widetilde{g}_j^{\log} h_j^{-1}[z, \zeta] = \Big[g_j z, R_{g_j}(z)\zeta - \Big(\frac{\ell(z)}{2\pi i} - \frac{1}{m_j} \Big) \Pi(g_j z) v_j \Big].$$

But

$$\frac{\ell'(g_j z)}{2\pi i} = \frac{\ell(z)}{2\pi i} - \frac{1}{m_j},$$

so applying h_j cancels the displayed translation and proves the global identity

$$h_j \widetilde{g}_j^{\log} h_j^{-1} = \widetilde{g}_j^0. \tag{2.9e}$$

Finally, by the stabilizer and translate conditions in (2.8b),

$$J|_{D_j^*} = (J_j'|_{\Delta_j^*})/\langle \widetilde{g}_j^0 \rangle.$$

Thus (2.9e) descends to the asserted punctured-disc biholomorphism. □

Lemma 6.6 (radial retraction and the finite-fibre stalk map)**.** ***[VERIFIED]*** *Under theorem 6.5, the finite local model N_j strongly deformation retracts onto its reduced central fibre S_j. After identifying a nearby fibre with the marked real torus $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$ by the flat coordinates Ξ, the specialization map*

$$H^r(S_j;\mathbb{Z}) \longrightarrow H^r(A_s;\mathbb{Z})$$

is the pullback π_j^* of the quotient covering

$$\pi_j : A_0^{(j)} \longrightarrow S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle.$$

Proof. Use the linearizing coordinate to define the radial contraction $r_u : \Delta_j \to \Delta_j$ by

$$s_j(r_u(z)) = (1-u)s_j(z), \qquad 0 \le u \le 1.$$

Because $s_j \circ g_j = \zeta_j s_j$, this contraction commutes with g_j. In the global real-analytic marking

$$\Xi(z, x+\Lambda) = [z, \Pi(z)x],$$

put

$$H_u\big(\Xi(z, x+\Lambda)\big) = \Xi(r_u(z), x+\Lambda).$$

Formula (2.8f) shows directly that H_u commutes with the logarithmic lift: it keeps x fixed while both composites send it to $A_j x + v_j/m_j$. Hence H_u descends to a strong deformation retraction of N_j onto S_j.

Fix $t_j \neq 0$ and one root s with $s^{m_j} = t_j$. Every orbit of the logarithmic lift meets the fibre over s exactly once, so that fibre maps isomorphically onto $f_j^{-1}(t_j)$. Following the descended radial retraction sends the marked point $x \in A_s$ to the orbit of the same x in the central torus, namely $\pi_j(x)$. The induced map from central-fibre cohomology to nearby-fibre cohomology is therefore exactly π_j^*, with no unrecorded conjugacy or index. □

Theorem 6.7 (global analytic construction)**.** ***[VERIFIED]*** *Gluing the period family over B° to N_0, N_1, N_2 by the preceding punctured-disc biholomorphisms produces a connected compact Hausdorff complex threefold X and a proper surjective holomorphic map*

$$f : X \longrightarrow B \simeq \mathbb{P}^1$$

with connected fibres. The fibre over p_0 is exactly the reduced nonnormal surface W of theorem 6.3; those over p_1, p_2 have scheme multiplicities three and four.

Proof. Take the disjoint union of the four manifolds and identify the two copies of each punctured-disc restriction by the displayed biholomorphism. The special discs are pairwise disjoint, so every nonsingleton equivalence class has two points and meets each summand at most once. The saturation of an open set is the union of that set with its images under finitely many overlap biholomorphisms, hence is open. Each summand therefore embeds openly in the quotient, and the only transition maps are biholomorphisms. This gives a complex threefold.

Points over distinct base values have disjoint disc preimages. Points over one base value lie in a common one of the four Hausdorff charts, so the quotient is Hausdorff. The local maps to the base agree on overlaps. Each is proper; properness is local on a locally compact Hausdorff base, so the global map is proper. Since the base is compact, $X = f^{-1}(B)$ is compact. All four pieces are connected and every local filling meets the central period-family piece, so X is connected. The fibre statements follow from theorems 6.3 and 6.5; their fibres are connected, and surjectivity follows. □

What has and has not been discharged. The formulas above reconstruct the period functions, both finite fillings, the toric cusp quotient, and all three punctured-overlap biholomorphisms. Thus theorems 6.3 and 6.7 discharge existence, smoothness, properness, the full central-fibre equation, and the normalization assertions formerly labelled (H1)–(H3) in theorem 11.10. This promotion is local to the analytic construction: no collapse, Leray transgression, or Mayer–Vietoris computation is used here. Integral cohomology and sphere recognition remain separate below.

7 Integral topology without the cusp-collapse argument

This section follows the independent nearby-cycle/Leray route. It does not use the candidate's cellular collapse, its Mayer–Vietoris homology matrix, or the asserted equality between those two constructions. All coefficient groups are integral.

Put

$$B = \mathbb{P}^1, \quad \Sigma = \{p_0, p_1, p_2\}, \quad B^\circ = B \setminus \Sigma, \quad j : B^\circ \hookrightarrow B,$$

and let F be a smooth fibre. Retain

$$V = H^1(F) = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \quad q = uw + 6\gamma\delta, \quad \mathrm{vol} = \gamma uw\delta.$$

7.1 Integral specialization at the cusp

Theorem 7.1 (nearby-cycle specialization)**.** ***[VERIFIED]*** *For the proper holomorphic cusp filling $f_0 : N_0 \to \Delta$ of theorem 6.3, with its proved local normal-crossings equations and normalization combinatorics, the specialization map of the central fibre W,*

$$\mathrm{sp}^r : H^r(W;\mathbb{Z}) \longrightarrow H^r(F;\mathbb{Z})$$

is injective and

$$\mathrm{im}\,\mathrm{sp}^r = H^r(F;\mathbb{Z})^{T_0} \tag{5.1}$$

for $0 \le r \le 4$. In particular, the image is saturated and there is no unrecorded finite-index defect at p_0.

Proof. Let $\widetilde{\Delta}^* \to \Delta^*$ be the universal cover, put $\widetilde{N}_0^* = N_0 \times_\Delta \widetilde{\Delta}^*$, and let

$$k : \widetilde{N}_0^* \longrightarrow N_0, \qquad i : W \hookrightarrow N_0$$

be the natural maps. The analytic nearby-cycle complex used here is

$$\psi = \psi_{f_0}\mathbb{Z} := i^* R k_* k^* \mathbb{Z}_{N_0} \in D_c^b(W). \tag{5.0}$$

Dimca defines analytic nearby cycles in [23, Definition 4.2.1]; the Milnor-fibre description of their stalks is [23, Proposition 4.2.2]. The proper-direct-image theorem for nearby cycles is [23, Proposition 4.2.11]; applied to f_0 and the identity map of the disc, it gives

$$\mathbb{H}^n(W;\psi) \simeq H^n(F;\mathbb{Z})$$

and identifies the morphism induced by the unit $\mathbb{Z}_W \to \psi$ with specialization. Proper base change itself is [23, Theorem 2.3.26]. Dimca's coefficient hypothesis is a commutative noetherian ring of finite global dimension, which $\mathbb{Z}$ satisfies. Thus the general integral analytic formalism is available here. The candidate-specific existence and local description of f_0 are proved in theorems 6.3 and 6.4. Kashiwara–Schapira [24, §8.6] corroborates the analytic formalism, while SGA 7, Exposé XIII [22], is its algebraic antecedent.

At a point where $f = z_0 \cdots z_k$, $k = 0, 1, 2$, the Milnor fibre deformation retracts onto $(S^1)^k$: in polar coordinates the radial variables form a contractible open simplex and the phases satisfy one linear relation. Hence

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \qquad \mathcal{H}^1\psi =: \mathscr{V}, \qquad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q, \qquad \mathcal{H}^b\psi = 0 \ (b \geq 3), \tag{5.2}$$

where $\mathscr{V}$ is supported on the three double curves $\bar{C}_1, \bar{C}_2, \bar{C}_3 \simeq \mathbb{P}^1$.

Orient each vanishing circle by the displayed order of its two branches in the normalization. At P, the branches of $(\bar{C}_1, \bar{C}_2, \bar{C}_3)$ are respectively

$$(b_0, b_1), \qquad (b_0, b_2), \qquad (b_1, b_2).$$

The three restrictions are therefore evaluations on

$$e_0 - e_1, \qquad e_0 - e_2, \qquad e_1 - e_2,$$

whose alternating sum is zero. At Q, the ordered branch pairs are

$$(c_2, c_1), \qquad (c_0, c_1), \qquad (c_0, c_2),$$

so the evaluations are

$$e_2 - e_1, \qquad e_0 - e_1, \qquad e_0 - e_2,$$

and again the first minus the second plus the third is zero. The global normalization labels C_k and C_{k+3} distinguish the two branches of each double curve and therefore fix these signs simultaneously. Thus

$$0 \longrightarrow \mathscr{V} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0,$$

$$\delta_1(n_1, n_2, n_3) = (s, s), \qquad s = n_1 - n_2 + n_3. \tag{5.3}$$

The image $\mathbb{Z}(1, 1)$ is primitive. Taking cohomology of (5.3) gives free groups with ranks

$$\operatorname{rk} H^a(W; \mathscr{V}) = (2, 1, 3, 0) \quad (a = 0, 1, 2, 3). \tag{5.4}$$

The normalization resolution for the constant sheaf is

$$0 \longrightarrow \mathbb{Z}_W \longrightarrow \nu_*\mathbb{Z}_{\mathrm{dP}_6} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0. \tag{5.5}$$

This is exact stalkwise. At a smooth point it is the identity $\mathbb{Z} \to \mathbb{Z}$. At a generic double point it is the augmented complex

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^2 \xrightarrow{(x_0, x_1) \mapsto x_0 - x_1} \mathbb{Z} \longrightarrow 0.$$

At a triple point, in either of the branch orders just fixed, it is

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^3 \xrightarrow{d_0} \mathbb{Z}^3 \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0,$$

where at P

$$d_0(x_0, x_1, x_2) = (x_0 - x_1, x_0 - x_2, x_1 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3,$$

while at Q

$$d_0(x_0, x_1, x_2) = (x_2 - x_1, x_0 - x_1, x_0 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3.$$

These augmented simplicial complexes are exact, which proves (5.5), including its endpoint signs.

On H^2, the only nontrivial map is

$$d : H^2(\mathrm{dP}_6;\mathbb{Z}) \longrightarrow \mathbb{Z}^3, \qquad d(L) = (L\cdot C_k - L\cdot C_{k+3})_{k=1}^3.$$

Write dP_6 as the blow-up of $\mathbb{P}^2$ at three noncollinear points, with basis H, E_1, E_2, E_3, and order its hexagon as

$$E_1, H-E_1-E_2, E_2, H-E_2-E_3, E_3, H-E_1-E_3.$$

For $L = aH - \sum b_i E_i$, direct intersection gives

$$d(L) = (s,-s,s), \qquad s = b_1+b_2+b_3-a. \tag{5.6}$$

Thus $\operatorname{im} d = \mathbb{Z}(1,-1,1)$ is primitive, $\ker d \simeq \mathbb{Z}^3$, and $\operatorname{coker} d \simeq \mathbb{Z}^2$. The resolution (5.5) has width two and the cohomology of its smooth strata is concentrated in even degrees. The only possible higher differential would be $d_2 : E_2^{0,q} \to E_2^{2,q-1}$, whose target is an odd row and is zero; differentials with $r \geq 3$ leave the width-two range. Therefore

$$H^a(W;\mathbb{Z}) = (\mathbb{Z},\mathbb{Z}^2,\mathbb{Z}^4,\mathbb{Z}^2,\mathbb{Z}) \quad (a=0,1,2,3,4), \tag{5.7}$$

with no torsion.

The hypercohomology spectral sequence

$$E_2^{a,b} = H^a(W;\mathcal{H}^b\psi) \Longrightarrow H^{a+b}(F;\mathbb{Z}) \tag{5.8}$$

has the following ranks, all on free groups:

$b=2$	2	0	0	0	0
$b=1$	2	1	3	0	0
$b=0$	1	2	4	2	1
	$a=0$	$a=1$	$a=2$	$a=3$	$a=4.$

Let the possible differential ranks be

$$\alpha = \operatorname{rk} d_2^{0,1}, \quad \beta = \operatorname{rk} d_2^{1,1}, \quad \gamma' = \operatorname{rk} d_2^{2,1}, \quad \delta' = \operatorname{rk} d_2^{0,2}, \quad \epsilon = \operatorname{rk} d_3^{0,2}.$$

Comparison with the torus Betti numbers $(1,4,6,4,1)$ in total degrees one, four, and two gives

$$\alpha = 0, \qquad \gamma' = 0, \qquad \beta+\delta'+\epsilon = 1. \tag{5.9}$$

We determine the remaining ambiguity geometrically, keeping the supported specialization map explicit. Choose a compact T_f-orbit $K = T_f\cdot w_0 \simeq T^2$ in the smooth stratum $W^\circ \simeq (\mathbb{C}^*)^2$. On a relatively compact neighbourhood of K, average a Riemannian metric over T_f and horizontally lift $\partial_{\operatorname{Re} t}$ and $\partial_{\operatorname{Im} t}$. If ξ_1^h, ξ_2^h are those lifts, put

$$\xi_t = (\operatorname{Re} t)\xi_1^h + (\operatorname{Im} t)\xi_2^h.$$

Choose a relatively compact T_f-invariant open neighbourhood K' with $K \Subset K'$. For some $r_0 > 0$, the time-one flow

$$\Theta : K' \times \Delta_{r_0} \longrightarrow N_0, \qquad \Theta(w,t) = \Phi_1^{\xi_t}(w), \tag{5.9a}$$

is a T_f-equivariant open embedding over Δ_{r_0}, with $\Theta(w,0) = w$. Indeed $df_0(\xi_t) = t$, so $f_0(\Theta(w,t)) = t$; its differential is invertible along $K'\times\{0\}$, and injectivity follows first by applying f_0 and then from uniqueness for the time-one flow. Shrinking K' and r_0 gives the claimed embedding.

For $0 < r < r_0$, set

$$A_r = f_0^{-1}(\Delta_r), \qquad Z_r = \Theta(K\times\Delta_r), \qquad K_t = \Theta(K\times\{t\}).$$

The subset Z_r is closed in A_r. Indeed, if a sequence in Z_r converges in A_r, compactness of K gives a convergent subsequence of its K-coordinates; its base coordinates converge to the image under f_0, which still lies in Δ_r, and the embedding Θ then identifies the limit with a point of Z_r.

For $0 < |t| < r$, functoriality of relative cohomology gives the two commutative squares

$$\begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_0\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_0} \\ H^3(W, W \setminus K) & \longrightarrow & H^3(W) \end{array} \qquad \begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_t\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_t} \\ H^3(F_t, F_t \setminus K_t) & \longrightarrow & H^3(F_t). \end{array} \tag{5.9b}$$

The two left vertical arrows are isomorphisms. By excision they are the restrictions to the two slices of

$$H^3\big(K' \times \Delta_r, (K' \setminus K) \times \Delta_r\big),$$

and homotopy invariance makes both restrictions isomorphisms. Taking the direct limit over shrinking r, the maps ρ_0 induce the proper-base-change isomorphism

$$\varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\sim} H^3(W; \mathbb{Z}).$$

Choose the points t on one radial ray and identify the F_t with the fixed marked fibre F by parallel transport. Restriction to F_t, followed by this identification, is compatible under shrinking r. The nearby-cycle specialization is precisely the composite

$$\mathrm{sp}^3 : H^3(W; \mathbb{Z}) \xleftarrow[\sim]{\rho_0} \varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\rho_{\mathrm{near}}} H^3(F; \mathbb{Z});$$

this is the neighbourhood realization of the unit $\mathbb{Z}_W \to \psi_{f_0}\mathbb{Z}$ in the comparison cited above. Taking the direct limit in the two functorial squares therefore proves, rather than assumes, that

$$H^3(W, W \setminus K) \longrightarrow H^3(W) \xrightarrow{\mathrm{sp}^3} H^3(F_t) \tag{5.9c}$$

is the transported supported class followed by the Gysin map for $K_t \hookrightarrow F_t$.

Under the marking of the torus fibre,

$$\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$$

is a saturated direct summand of Λ, and K_t is a translate of $(\Lambda_{\mathrm{tor}} \otimes \mathbb{R})/\Lambda_{\mathrm{tor}}$. Choose a complementary summand Λ', and write $F_t \simeq T_f \times T_B$ as oriented real tori, where $T_B = (\Lambda' \otimes \mathbb{R})/\Lambda'$, with the product orientation chosen to agree with that of F_t. The real rank-two normal bundle of the translate K_t is trivial and oriented. The Thom isomorphism gives

$$H^3(F_t, F_t \setminus K_t) \simeq H^1(T_f),$$

and its map to absolute cohomology is

$$\iota_!(a) = a \otimes [T_B]^\vee \in H^1(T_f) \otimes H^2(T_B) \subset H^3(F_t). \tag{5.9d}$$

This is an injective primitive rank-two Künneth direct summand. On the central smooth stratum, $W^\circ \simeq (\mathbb{C}^*)^2 \simeq T_f \times \mathbb{R}^2$ as an oriented real manifold; hence the real rank-two normal bundle of K is also trivial and oriented. The same Thom calculation gives $H^3(W, W \setminus K) \simeq H^1(K) \simeq \mathbb{Z}^2$. Hence (5.9c) has primitive rank-two image. Since $\mathrm{rk}\, H^3(W) = 2$, specialization in degree three is injective and has precisely that primitive image. In (5.9), this forces $\beta = \epsilon = 0$, and consequently $\delta' = 1$.

Specialization is now injective in every degree. Its image is primitive: in degrees one and two the quotient filtration in (5.8) has free graded pieces; degree three was just proved; and in degree four the map is an isomorphism. Monodromy invariance gives

$$\operatorname{im} \mathrm{sp}^r \subseteq H^r(F;\mathbb{Z})^{T_0}.$$

By theorem 5.2, the invariant ranks are $(1,2,4,2,1)$, exactly those in (5.7), and each invariant lattice is the kernel of an integral endomorphism, hence primitive. Two primitive sublattices of the same rational subspace coincide: both equal the intersection of that subspace with the ambient lattice. This proves (5.1). □

7.2 The global E_2-page

The Leray spectral sequence used in this subsection is [23, Corollary 2.3.4]. Let $R^r = R^r f_*\mathbb{Z}$ and $\mathcal{L}^r = j^*R^r$. The finite-fibre lattices have already been proved as abstract lattice statements in theorem 5.4. Their identification with the stalks of the proposed family requires the explicit radial-stalk compatibility below. This compatibility is treated separately in theorem 6.6 and is not silently absorbed here.

Proposition 7.2 (all Leray terms used below)**.** ***[VERIFIED]*** *The specialization theorem theorem 7.1, the finite fillings of theorem 6.5, and theorem 6.6 give, for $j = 1,2$ and $0 \le r \le 4$, the finite-filling compatibility*

$$N_j \text{ strongly deformation retracts onto } S_j, \qquad H^r(S_j;\mathbb{Z}) \longrightarrow H^r(F;\mathbb{Z})^{T_j} \text{ is } \pi_j^* \tag{FF}$$

under the marked punctured-fibre identification. Consequently,

$$H^0(B,R^1) = \mathbb{Z}(12\gamma), \quad H^0(B,R^2) = \mathbb{Z}(2q), \quad H^0(B,R^3) = \mathbb{Z}(2\gamma uw), \tag{5.10}$$

$$H^1(B,R^1) = H^1(B,R^2) = 0, \tag{5.11}$$

and, with ω a generator of $H^2(B;\mathbb{Z})$,

$$H^2(B,R^1) = \mathbb{Z}\omega[\delta], \qquad H^2(B,R^2) = \mathbb{Z}\omega[\gamma\delta], \qquad H^2(B,R^4) = \mathbb{Z}\omega[\mathrm{vol}]. \tag{5.12}$$

These identifications are compatible with exterior products in V.

Proof. Under the stated hypotheses, proper base change identifies the stalk of R^r at p_0 with $H^r(W;\mathbb{Z})$, while $(j_*\mathcal{L}^r)_{p_0} = H^r(F;\mathbb{Z})^{T_0}$. The stalk map is specialization and is an isomorphism by theorem 7.1; hence its cokernel at p_0 is zero. At p_j, condition (FF) identifies the stalk with $H^r(S_j;\mathbb{Z})$, the target with $H^r(F;\mathbb{Z})^{T_j}$, and the stalk map with π_j^*. The finite-fibre calculation theorem 5.4 makes each of these maps injective. Away from the three special points the map is the identity of the local system. Stalkwise kernels and cokernels therefore give the exact sequence

$$0 \longrightarrow R^r \longrightarrow j_*\mathcal{L}^r \longrightarrow Q^r \longrightarrow 0, \tag{5.13}$$

with

$$Q^1 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}, \quad Q^2 = (\mathbb{Z}/2)_{p_2}, \quad Q^3 = (\mathbb{Z}/2)_{p_2}, \quad Q^4 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}.$$

The degree-one quotients have orders three and four and are generated by the class of γ. In degree two the p_1-quotient is zero, while

$$(\textstyle\bigwedge^2 V)^{T_2} / \{a\gamma\eta_2 + bq : a \equiv b \pmod 2\} \simeq \mathbb{Z}/2$$

is generated by the class of q. In degree three the p_1-quotient is zero and the p_2-quotient has order two, generated by the class of γuw. This last identification uses the actual twist $v_2 = -\varepsilon'$, for which

the source's admissibility quantity $\eta_2(v_2)$ is zero and hence divisible by four. In degree four, pullback multiplies the orientation class by the covering degree m_j, giving the displayed Q^4. Hence the skyscraper sheaves have exactly the asserted stalks; no claim for an arbitrary twist is being imported. By theorem 5.3,

$$V^G = \mathbb{Z}\gamma, \quad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \quad (\textstyle\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \tag{5.14}$$

and

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad [q] = 12[\gamma\delta]. \tag{5.15}$$

Since $\det T_1 = \det T_2 = 1$, the action on $\bigwedge^4 V = \mathbb{Z}\,\mathrm{vol}$ is trivial and

$$(\textstyle\bigwedge^4 V)_G = \mathbb{Z}[\mathrm{vol}]. \tag{5.15a}$$

The maps from the invariant groups in (5.14) onto the corresponding skyscraper fibres are surjective, with kernels $12\mathbb{Z}\gamma$, $2\mathbb{Z}q$, and $2\mathbb{Z}\gamma uw$. This proves (5.10).

Because each Q^r is a skyscraper, it has no positive cohomology; the long exact sequence of (5.13) therefore identifies $H^2(B, R^r)$ with $H^2(B, j_*\mathcal{L}^r)$. For any free local system $\mathcal{L}$ on B°, with stalk M, the quotient $j_*\mathcal{L}/j_!\mathcal{L}$ is a skyscraper. Hence

$$H^2(B, j_*\mathcal{L}) \simeq H^2_c(B^\circ, \mathcal{L}) \simeq H_0(B^\circ, \mathcal{L}) = M_G, \tag{5.16}$$

where the middle isomorphism is Poincaré duality with local coefficients [9, §3.H]; see also G. E. Bredon, *Sheaf Theory*, 2nd ed., Chapter VI. Cap product with the locally finite fundamental class makes (5.16) natural for coefficient pairings, proving (5.12) and the product assertion.

It remains to prove (5.11), for which we spell out the parabolic-cohomology quotient. The low-degree Leray sequence for $j : B^\circ \hookrightarrow B$ contains

$$0 \longrightarrow H^1(B, j_*\mathcal{L}) \longrightarrow H^1(B^\circ, \mathcal{L}) \longrightarrow H^0(B, R^1 j_*\mathcal{L}).$$

If $I_i = \langle t_i \rangle$ is the local inertia group at p_i, then

$$(R^1 j_*\mathcal{L})_{p_i} = H^1(I_i, M) = M/(T_i - I)M.$$

Consequently the last arrow sends the class of a cocycle c to the three classes of $c(t_i)$, and $H^1(B, j_*\mathcal{L})$ is exactly the kernel defined by the three parabolic conditions below. Use left cocycles, so $c(gh) = c(g) + gc(h)$. A cocycle on $\pi_1(B^\circ) = F(t_1, t_2)$ is determined by $x = c(t_1)$ and $y = c(t_2)$. Extension across the three punctures is the parabolic condition

$$x \in (T_1 - I)M, \qquad y \in (T_2 - I)M, \qquad c(t_0) \in (T_0 - I)M, \qquad t_0 = (t_1 t_2)^{-1}.$$

Write $y = (T_2 - I)m$ and subtract the coboundary of m; this normalizes $y = 0$. A further coboundary preserves the normalization precisely when its defining element lies in M^{T_2}, and then changes x by an element of $(T_1 - I)M^{T_2}$. With $y = 0$, $c(t_0) = -T_0 x$. Since T_0 preserves $(T_0 - I)M$, the last parabolic condition is equivalent to $x \in (T_0 - I)M$. Therefore

$$H^1(B, j_*\mathcal{L}) \simeq \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}. \tag{5.17}$$

For $M = V$,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u \rangle, \qquad (T_0 - I)V = \langle \gamma, u \rangle,$$

and their intersection is $\mathbb{Z}(2\gamma - u) = (T_1 - I)\mathbb{Z}\eta_2$. For $M = \bigwedge^2 V$,

$$(\textstyle\bigwedge^2 T_1 - I)\bigwedge^2 V = \langle \gamma u, \gamma w, 2u\delta + w\delta, \\ uw + u\delta - w\delta - 6\gamma\delta \rangle,$$

whereas

$$(\textstyle\bigwedge^2 T_0 - I)\bigwedge^2 V = \langle \gamma u, \gamma w + u\delta \rangle.$$

Their intersection is $\mathbb{Z}\gamma u$, equal to $(\bigwedge^2 T_1 - I)\mathbb{Z}(\gamma\eta_2)$. Thus both groups in (5.17) vanish. The maps $H^0(B, j_*\mathcal{L}^r) \to H^0(B, Q^r)$ were proved surjective above for $r = 1, 2$, and skyscraper sheaves have no positive cohomology. The long exact sequence of (5.13) therefore gives (5.11) for R^1, R^2. □

7.3 The three transgressions

Lemma 7.3 (the primitive first transgression)**.** ***[VERIFIED]*** *For the constructed map and the based meridians of theorem 7.6, one may orient ω so that*

$$d_2^{0,1}(12\gamma) = \omega. \tag{5.18}$$

Proof. Choose disjoint discs D_i about the special points and a pair-of-pants neighbourhood V_0 of their complement. The class $\theta = 12\gamma$ is a section of $R^1 f_*\mathbb{Z}$ over each D_i: at p_0 this is theorem 7.1, and at p_j it is the degree-one finite-fibre lattice. It is also a section over V_0, because it is globally monodromy invariant. For every member $U \in \{D_0, D_1, D_2, V_0\}$, the local Leray five-term sequence contains

$$H^1(f^{-1}(U);\mathbb{Z}) \longrightarrow H^0(U, R^1 f_*\mathbb{Z}) \longrightarrow H^2(U, R^0 f_*\mathbb{Z}) = H^2(U;\mathbb{Z}) = 0.$$

Thus the section θ has a cohomological lift $\Theta_U \in H^1(f^{-1}(U);\mathbb{Z})$ on every member of the cover. On an overlap $D_i \cap V_0$, the two lifts have the same image in $H^0(R^1 f_*\mathbb{Z})$; local five-term exactness therefore identifies their difference with a class in $H^1(D_i \cap V_0, R^0 f_*\mathbb{Z}) = H^1(D_i \cap V_0;\mathbb{Z})$. These three differences form the Čech connecting cocycle used below.

Use clockwise meridians. The affine deck relation at p_j is

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.19}$$

Put $\Theta_j = \Theta_{D_j}$. Since Θ_j is integral and restricts to θ on the fibre, (5.19) gives $m_j\Theta_j(\widehat{\rho}_j) = \Theta_j(t_{v_j}) = \theta(v_j)$. Therefore

$$\Theta_1(\widehat{\rho}_1) = 12\gamma(v_1)/3 = 4, \qquad \Theta_2(\widehat{\rho}_2) = 12\gamma(v_2)/4 = -3. \tag{5.20}$$

By theorem 7.6, the toric meridian $\widetilde{a}_0$ is the boundary of the holomorphic disc $\{(1,1,t) : |t| \le \epsilon\}$. It is therefore null-homotopic and the local lift evaluates to zero.

Write $\alpha_i = \Theta_{V_0}(\widehat{\rho}_i)$. By the exact based cusp relation (5.23a), the three annular discrepancies have meridian values

$$4 - \alpha_1, \qquad -3 - \alpha_2, \qquad \alpha_1 + \alpha_2, \tag{5.21}$$

whose sum is one. Mayer–Vietoris on the base identifies

$$H^1\left(\coprod_i (D_i \cap V_0)\right) / \operatorname{im} H^1(V_0) \simeq H^2(B;\mathbb{Z}).$$

Under meridian evaluation the numerator is $\mathbb{Z}^3$, the image is the sum-zero lattice, and the connecting map is the signed sum of the three coordinates.

We now identify this connecting class with the Leray differential rather than taking that identification as an extra hypothesis. Put $K = Rf_*\mathbb{Z}_X$. The canonical truncation triangle [26, Tag 08J5] is

$$R^0 f_*\mathbb{Z}_X \longrightarrow \tau_{\le 1} K \longrightarrow R^1 f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0 f_*\mathbb{Z}_X[1]. \tag{5.21a}$$

After applying $R\Gamma(B, -)$, the connecting homomorphism

$$H^0(B, R^1 f_*\mathbb{Z}) \xrightarrow{H^0(\kappa)} H^2(B, R^0 f_*\mathbb{Z}) \tag{5.21b}$$

is precisely $d_2^{0,1}$ in the hypercohomology, hence Leray, spectral sequence; this is the canonical-truncation construction of that spectral sequence [26, Tag 0FLJ]. On the chosen disc–pair-of-pants cover, functorial Mayer–Vietoris for the triangle [26, Tag 0F16] computes $H^0(\kappa)$ by choosing the local degree-one lifts Θ_U and taking their differences on the annular overlaps. Those differences are exactly the cocycle (5.21). Thus the signed sum just computed is $d_2^{0,1}(\theta)$. Equation (5.21) therefore represents a primitive generator; orient ω to obtain (5.18). □

Lemma 7.4 (the higher transgressions)**.** ***[VERIFIED]*** *In the integral Leray spectral sequence of the constructed map,*

$$d_2^{0,2}(2q) = \omega[\delta], \qquad d_2^{0,3}(2\gamma uw) = \omega[\gamma\delta]. \tag{5.22}$$

Proof. Use the multiplicative Leray spectral sequence obtained from a filtered multiplicative cochain resolution. Its E_2-product is the natural cup product

$$H^p(B, R^q) \otimes H^{p'}(B, R^{q'}) \longrightarrow H^{p+p'}(B, R^{q+q'}),$$

and every d_r is a graded derivation. This is the standard filtered-DGA construction; compare the filtered-complex spectral sequence [26, Tags 012K and 0FLJ] with the derived cup product [26, §20.31]. The generator identifications and their products are natural by the last sentence of theorem 7.2. Put $\theta = 12\gamma$, $a = 2q$, and $b = 2\gamma uw$. By theorem 7.2, write

$$d_2 a = r\omega[\delta], \qquad d_2 b = s\omega[\gamma\delta].$$

In the E_2-ring,

$$\theta a = 12b, \qquad ab = 0, \qquad [q] = 12[\gamma\delta].$$

The derivation rule and $|\theta| = 1$ give

$$12 d_2 b = d_2(\theta a) = d_2\theta \cdot a - \theta \cdot d_2 a,$$

hence $s = 2 - r$. Since $|a| = 2$, $0 = d_2(ab) = d_2 a \cdot b + a \cdot d_2 b$. The exterior products are

$$\delta \wedge 2\gamma uw = -2\,\mathrm{vol}, \qquad 2q \wedge \gamma\delta = 2\,\mathrm{vol}\,.$$

Therefore $-2r + 2s = 0$, so $r = s$. Together with $s = 2 - r$, this gives $r = s = 1$, proving (5.22). □

Theorem 7.5 (integral cohomology in low degrees)**.** ***[VERIFIED]*** *For the constructed holomorphic map,*

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0. \tag{5.23}$$

Proof. The Leray terms in total degree two are

$$E_2^{2,0} = \mathbb{Z}\omega, \qquad E_2^{1,1} = 0, \qquad E_2^{0,2} = \mathbb{Z}(2q).$$

By theorems 7.3 and 7.4, the incoming map to the first term and the outgoing map from the last term are isomorphisms. Thus all three E_∞-pieces vanish and $H^2(X) = 0$. The first map has zero kernel, giving $H^1(X) = 0$. In total degree three, the quotient of $E_2^{2,1}$ by the image of $E_2^{0,2}$ is zero, $E_2^{1,2} = 0$, and the outgoing map from $E_2^{0,3}$ is injective by (5.22). Hence $H^3(X) = 0$. No later differential can meet these terms because the base has cohomological dimension two.

The universal coefficient sequences show both $\mathrm{Hom}(H_2(X), \mathbb{Z}) = 0$ and $\mathrm{Ext}(H_2(X), \mathbb{Z}) = 0$. Since $H_2(X)$ is finitely generated, it has neither a free nor a torsion summand, so it is zero. □

7.4 Based van Kampen and sphere recognition

Proposition 7.6 (the geometric fundamental-group presentation)**.** ***[VERIFIED]*** *For the punctured-overlap gluings of theorems 6.4 and 6.5, the based van Kampen presentation is*

$$\pi_1(X) \simeq \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1. \tag{5.24}$$

Proof. Choose b_* in a pair-of-pants core of B°, put x_* on the holomorphic zero section above it, and choose three nonintersecting base whiskers from b_* to clockwise boundary circles. Lift all whiskers by the zero section. With the ordering compatible with $g_1g_2g_0 = 1$, the resulting based loops obey

$$\widehat{\rho}_1\widehat{\rho}_2\widetilde{a}_0 = 1. \tag{5.23a}$$

The period-family action is linear and hence preserves the zero section. It splits the homotopy sequence over the pair of pants, giving

$$\pi_1(f^{-1}(B^\circ)) = \Lambda \rtimes F(\widehat{\rho}_1, \widehat{\rho}_2),$$

with conjugation actions A_1, A_2. The clockwise meridian at p_j lifts to the affine generator in (2.9), including its sign and based translation. Indeed, on a logarithm branch put $\alpha = \log(s_j)/(2\pi i)$. The overlap map h_j translates flat coordinates by $+\alpha v_j$, so the inverse image of the period-family zero section is $x(z) = -\alpha(z)v_j$. Clockwise continuation takes z to g_jz and $\alpha(g_jz) = \alpha(z) - 1/m_j$. Because $A_jv_j = v_j$, the affine generator sends the initial point to

$$A_j(-\alpha v_j) + v_j/m_j = -(\alpha - 1/m_j)v_j,$$

exactly the continued endpoint. Thus the based zero-section meridian is the affine generator, not its inverse or a translated conjugate. Formula (2.8f) gives

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.23b}$$

The radial model of theorem 6.6 shows that attaching N_j imposes exactly this relation. Since the global and local whiskers are the same zero-section path under the overlap, conjugating back to x_* adds no fibre drift.

For completeness, the cusp covering has no hidden fundamental group. A loop in the infinite toric variety has compact image and therefore lies in the open $X_{\Sigma_{\mathrm{tor}}}$ associated with a finite subfan Σ_{tor}. Enlarge that subfan by the rays through $(0,1), (e_1,1), (e_2,1)$. Write

$$N_{\Sigma_{\mathrm{tor}}} := \langle |\Sigma_{\mathrm{tor}}| \cap N' \rangle_{\mathbb{Z}}.$$

The three displayed primitive generators span N', so the classical formula $\pi_1(X_{\Sigma_{\mathrm{tor}}}) = N'/N_{\Sigma_{\mathrm{tor}}}$ [17, Theorem 12.1.10] makes the resulting finite toric open simply connected; the original loop therefore has a null-homotopy in that open. Thus Y is simply connected. The one-parameter subgroup in the third torus coordinate acts by $t \mapsto rt$, $r > 0$. Given the compact image of a null-homotopy, choose r so small that its translate lies in $|t| < \epsilon$; the path $r_s = (1-s) + sr$ gives a homotopy from the original loop to its translate inside Y_ϵ. Hence Y_ϵ is also simply connected. The verified free quotient $N_0 = Y_\epsilon/\bar{\Lambda}$ therefore has universal cover Y_ϵ, so $\pi_1(N_0) = \bar{\Lambda}$. Equation (2.8iii) identifies a fibre-lattice loop $\lambda \in \Lambda$ with the deck transformation $\bar{\lambda}$; hence the inclusion of a nearby fibre is exactly

$$\Lambda \longrightarrow \bar{\Lambda}, \qquad \lambda \longmapsto \bar{\lambda}, \tag{5.23c}$$

with kernel $\Lambda_{\mathrm{tor}} = \langle \widehat{w}, \widehat{\delta} \rangle$. On the zero section the same overlap sends $(z, 0)$ to $(1, 1, t_c)$. In the toric chart of the triangle $\{0, e_1, e_2\}$, whose coordinates may be written $(t_c/(x_1x_2), x_1, x_2)$, this is the holomorphic disc $t_c \mapsto (t_c, 1, 1)$. Thus the cusp meridian $\widetilde{a}_0$ is null-homotopic, with no fibre translation. Combining this fact with (5.23a), the cusp attachment imposes precisely $\widehat{\rho}_1\widehat{\rho}_2 = 1$.

Van Kampen now gives

$$\left\langle \Lambda, \widehat{\rho}_1, \widehat{\rho}_2 \;\middle|\; \widehat{\rho}_j\lambda\widehat{\rho}_j^{-1} = A_j\lambda, \widehat{\rho}_1^3 = t_{v_1}, \widehat{\rho}_2^4 = t_{v_2}, \widehat{\rho}_1\widehat{\rho}_2 = 1, \Lambda_{\mathrm{tor}} = 1 \right\rangle.$$

The normal closure of Λ_{tor} under A_1, A_2 is $\ker\gamma = \langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$; for example $A_1\widehat{w} = \widehat{u} - \widehat{w}$. The image of Λ is therefore generated by $c = [\widehat{\gamma}]$, which is central because γ is invariant. Since $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$, the presentation becomes (5.24). It is the case $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ of theorem 5.5, where $p = -1$; hence the group is trivial. □

Corollary 7.7 (topology and the third Chern number)**.** ***[VERIFIED]*** *The underlying smooth manifold of the constructed threefold is diffeomorphic to the standard six-sphere, and*

$$\langle c_3(T_X), [X] \rangle = 2. \tag{5.25}$$

Proof. Combine theorems 7.5 and 7.6. The universal coefficient theorem and Poincaré duality give the integral homology of S^6, exactly as in theorem 10.7. The complete recognition argument of theorem 10.8 then gives $X \cong_{\mathrm{diff}} S^6$. Its Euler characteristic is two. On a complex threefold the top Chern class of the complex tangent bundle is the Euler class of the underlying oriented real tangent bundle, proving (5.25). □

Remaining role of the collapse route. The verified non-collapse route above does not validate every geometric statement in the separate cellular-collapse/Mayer–Vietoris proof. Those statements remain worth auditing because the manuscript asserts the collapse map and detailed homology of W. They are no longer premises of the sphere-recognition conclusion: if their named analytic and geometric hypotheses are discharged, the integral nearby-cycle theorem and Leray calculation provide an independent proof.

8 The direct algebraic-reduction proof

The calculation in this section is independent of the integral topology of X. It uses the period family of theorem 6.1 and the proper connected fibration of theorem 6.7; it does not use $H^2(X;\mathbb{Z})$, the sphere-recognition argument, or any theorem of Campana–Demailly–Peternell.

Lemma 8.1 (algebraic reduction of a two-dimensional torus)**.** *Let T be a complex torus of dimension two. Suppose*

$$\mathrm{NS}(T) = \mathbb{Z}\eta, \qquad \eta^2 \neq 0,$$

and the Appell–Humbert Hermitian form H_η is nondegenerate and indefinite. Then $a(T) = 0$.

Proof. The algebraic reduction of a compact complex torus is a quotient $q : T \to A = T/K$ by a connected complex subtorus, with A an abelian variety and $\dim A = a(T)$ [14, Chapter 2, §6]; see also [15, §8, p. 2437].

Assume $a(T) > 0$. If $a(T) = 2$, then T is an abelian surface. An ample line bundle has first Chern class $n\eta$, $n \neq 0$, and its Appell–Humbert form nH_η is positive definite. This contradicts the indefiniteness of H_η.

If $a(T) = 1$, then A is an elliptic curve. For a point $p \in A$, put

$$\varphi = q^* c_1(\mathcal{O}_A(p)) \in \mathrm{NS}(T).$$

The class is nonzero: for any translation-invariant Kahler form ω on T, the projection formula gives $\int_T \varphi \wedge \omega = \int_K \omega > 0$. But $\varphi^2 = 0$, since A is a curve. Writing $\varphi = n\eta$, $n \neq 0$, yields $0 = n^2\eta^2 \neq 0$, a contradiction. Both possible positive algebraic dimensions are excluded. □

Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \eta = u \wedge w + 6\gamma \wedge \delta.$$

For an integral two-class $E \in \bigwedge^2 V$, write its coordinates in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$ as $E_{\gamma u}, \dots, E_{w\delta}$, and set

$$\begin{aligned} P_E = {} & E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta \\ & + E_{w\delta}(6\mu^2 - \tau\beta). \end{aligned} \tag{3.1}$$

Proposition 8.2 (very general Neron–Severi group)**.** ***[VERIFIED]*** *For the descended period family of theorem 6.1, outside a countable subset of B° the smooth fibre F_b satisfies*

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad a(F_b) = 0. \tag{3.2}$$

Proof. The two holomorphic one-forms defining the Hodge subspace are

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta.$$

The determinant inequality (2.2) gives $V_{\mathbb{C}} = F^1 \oplus \overline{F^1}$, where $F^1 = \langle \sigma_1, \sigma_2 \rangle$. For a real two-class E, the Hodge decomposition therefore gives

$$E \in H^{1,1}(F_b) \quad \Longleftrightarrow \quad E \wedge \sigma_1 \wedge \sigma_2 = 0.$$

Expanding the wedge product gives

$$E \wedge \sigma_1 \wedge \sigma_2 = P_E\, \gamma \wedge u \wedge w \wedge \delta. \tag{3.3}$$

We next determine exactly which P_E vanish identically. Suppose

$$a + b\tau + c\mu + d\beta + e(6\mu^2 - \tau\beta) \equiv 0. \tag{3.4}$$

Apply the cusp transformation $(\tau, \mu, \beta) \mapsto (\tau - 1, \mu, \beta + 1)$ and subtract. This gives

$$-b + d + e(\beta - \tau + 1) = 0.$$

Because $\beta - \tau$ is translated by two, $e = 0$, and then $d = b$. Apply the order-four transformation in (2.1), subtract (3.4), and multiply by $-\tau$. One obtains

$$6b\mu^2 + c(\tau - 1)\mu + b(\tau^2 + 3\tau + 1) - c\tau = 0.$$

Subtract its cusp translate:

$$c\mu + 2b(\tau + 1) - c = 0. \tag{3.5}$$

Translate (3.5) once around the cusp. Since μ is invariant and $\tau \mapsto \tau - 1$, subtracting the translated identity from (3.5) gives $2b = 0$. Thus $b = 0$, and (3.5) becomes $c(\mu - 1) = 0$. If $c \neq 0$, then $\mu \equiv 1$, whereas the order-four law would give $1 = \mu \circ g_2 = 1 + 1/\tau$, an impossibility. Hence $c = 0$, then $b = d = 0$, and finally $a = 0$. Thus the five functions

$$1, \quad \tau, \quad \mu, \quad \beta, \quad 6\mu^2 - \tau\beta$$

are linearly independent over $\mathbb{C}$.

Reading the coefficients in (3.1) now gives

$$P_E \equiv 0 \quad \Longleftrightarrow \quad E \in \mathbb{Z}\eta. \tag{3.6}$$

For each other integral class, the zero set of P_E on the connected Riemann surface $\mathfrak{h}_z$ is discrete, hence countable. There are only countably many integral classes. The class $\eta = q = uw + 6\gamma\delta$ is monodromy invariant by theorem 5.3, so it defines a class on every marked fibre after descent. The image in B° of the countable union of the remaining discrete zero sets is still countable. A base point outside that image has no additional integral $(1,1)$-class on any lift, proving the Neron–Severi statement on the quotient family.

Finally,

$$\eta^2 = 12\, \gamma \wedge u \wedge w \wedge \delta. \tag{3.7}$$

For a nondegenerate alternating form E, write $E^\vee$ for its inverse alternating form on the dual real vector space. If $G_\eta = (i\eta^\vee(\sigma_a, \overline{\sigma_b}))_{a,b}$, direct inversion of the alternating-form matrix and substitution give

$$G_\eta = \begin{pmatrix} 2\operatorname{Im}\tau & 2\operatorname{Im}\mu \\ 2\operatorname{Im}\mu & \frac{1}{3}\operatorname{Im}\beta \end{pmatrix},$$

$$\det G_\eta = \frac{2}{3}\operatorname{Im}\tau\left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}\right) < 0. \tag{3.8}$$

Here is the Appell–Humbert bridge needed to read the inertia of this matrix. With the convention $E = \operatorname{Im} H_E$, one has

$$E = \frac{1}{2i}\sum_{a,b}(H_E)_{ab}\,\sigma_a \wedge \overline{\sigma_b}.$$

Thus, in the ordered basis $(\sigma_1, \sigma_2, \overline{\sigma_1}, \overline{\sigma_2})$, the upper-right block of the alternating-form matrix of E is $A = -(i/2)H_E$. Inverting the block matrix gives

$$E^\vee(\sigma_a, \overline{\sigma_b}) = [-(A^t)^{-1}]_{ab} = -2i(\overline{H_E}^{-1})_{ab}, \qquad G_E = 2\overline{H_E}^{-1}. \tag{3.8a}$$

The displayed G_η is real symmetric, so $H_\eta = 2G_\eta^{-1}$. Inversion and multiplication by a positive scalar preserve inertia. Hence H_η is nondegenerate and indefinite, while (3.7) is nonzero. Apply theorem 8.1. □

Lemma 8.3 (meromorphic descent)**.** *Let $f : X \to B$ be a proper surjective holomorphic map from a connected complex threefold to a connected compact Riemann surface, with connected fibres, and a submersion over $B^\circ = B \setminus \Sigma$, where Σ is finite and $B^\circ \neq \varnothing$. If $g \in \mathcal{M}(X)$ is constant on the fibres over an uncountable subset of B°, then there is a unique $\varphi \in \mathcal{M}(B)$ such that $g = f^*\varphi$.*

Proof. Let P be the polar divisor of g, and put $\Omega = f^{-1}(B^\circ)\setminus|P|$. A proper analytic subset of a connected complex manifold has real codimension at least two and does not disconnect it. First note that $f^{-1}(B^\circ)$ is connected. Indeed, each of its connected components is open, and its image is open because f is a submersion and closed because the base change $f^{-1}(B^\circ) \to B^\circ$ is proper. Since B° is connected, every component would meet every fibre; connectedness of the fibres therefore permits only one component. Removing $|P|$ now leaves Ω connected. Only finitely many fibres are components of $|P|$. For every other b, the complement $F_b \cap \Omega$ of a proper analytic subset in the connected manifold F_b is connected.

On Ω, define the analytic set

$$Z = \{dg \wedge f^*(dt) = 0\},$$

where t is any local coordinate on B. If $g|_{F_b}$ is constant, then $F_b \cap \Omega \subset Z$. If $Z \neq \Omega$, then, because Ω is second countable, Z has at most countably many irreducible components. Each irreducible two-dimensional fibre contained in Z is one of those components, and distinct fibres give distinct components. This contradicts the uncountable hypothesis. Therefore $Z = \Omega$.

It follows that dg annihilates $\ker df$. Connectedness of $F_b \cap \Omega$, together with local holomorphic sections of the submersion, gives $g = \varphi \circ f$ on Ω, for a holomorphic φ away from finitely many points of B.

Fix such a point t_0. Choose a general holomorphic curve germ $\kappa : (\Delta, 0) \to (X, x)$, not contained in $|P|$, with

$$f \circ \kappa(s) = t_0 + s^k u(s), \qquad u(0) \neq 0.$$

If $g \circ \kappa$ has pole order at most M, then $|g \circ \kappa(s)| \leq C|s|^{-M}$. After shrinking the disc, boundedness above and below of $|u|$ gives constants $c, C_1 > 0$ such that

$$c|t - t_0|^{1/k} \leq |s| \leq C_1|t - t_0|^{1/k}$$

on each local inverse branch. Therefore

$$|\varphi(t)| \leq C_2|t - t_0|^{-M/k}$$

away from the image of a discrete exceptional set, and then everywhere on the punctured disc by holomorphy. For any integer $N > M/k$, the function $(t-t_0)^N\varphi(t)$ is bounded and extends holomorphically. Thus φ is meromorphic at t_0. Repeating at the finitely many exceptional points produces $\varphi \in \mathcal{M}(B)$. Equality on the dense open set Ω gives equality of meromorphic functions on X, and surjectivity of f gives uniqueness. □

Theorem 8.4 (algebraic dimension and algebraic reduction)**.** ***[VERIFIED]*** *For the global holomorphic family of theorem 6.7,*

$$\mathcal{M}(X) = f^*\mathcal{M}(\mathbb{P}^1) = f^*\mathbb{C}(t), \qquad a(X) = 1, \tag{3.9}$$

and f is the algebraic reduction.

Proof. By theorem 8.2, all but countably many smooth fibres have algebraic dimension zero. Let $g \in \mathcal{M}(X)$. Only finitely many fibres are components of its polar divisor, so on uncountably many of the preceding fibres the restriction of g is a meromorphic function on a torus of algebraic dimension zero and is therefore constant. By theorem 8.3, $g = f^*\varphi$ for a unique $\varphi \in \mathcal{M}(\mathbb{P}^1)$. The reverse inclusion is immediate, and (3.9) follows. Since f is proper, surjective, has connected fibres, and induces the equality of meromorphic function fields, it is the algebraic reduction. □

Disposition. The earlier label "algebraic reduction open" was too broad. The direct proof above validates the assertion independently of Section 7 and of the CDP comparison. Its provenance is the candidate's Lemmas 9.2–9.6 and Proposition 9.7, reconstructed here with an explicit Appell–Humbert matrix bridge, a proof that the smooth locus remains connected, and the omitted root estimate in the final meromorphic-extension step. This does not by itself prove the claimed topology of X; that is a separate calculation.

9 Structure-sheaf direct images and uniform non-torsion

This section reconstructs the analytic inputs used by the candidate's Proposition 10.8. The proof differs at one important point from the candidate's route: extension across the cusp is obtained from the intrinsic nearby-cycle isomorphism of theorem 7.1, not from the still-open comparison with the candidate's cellular collapse. Thus the result below belongs to the verified non-collapse route.

Put

$$e = d\zeta_1 \wedge d\zeta_2$$

on the marked torus family. Retain the local coordinates and notation of section 6:

$$s_j \circ g_j = \zeta_j s_j, \qquad (m_1, m_2) = (3, 4), \qquad t_j = s_j^{m_j}, \qquad f^{-1}(p_j) = m_j S_j.$$

9.1 The Hodge line and the finite normal characters

Lemma 9.1 (determinant characters and the Hodge line)**.** ***[VERIFIED]*** *Let*

$$\mathfrak{K} = f_*\omega_{X/B}.$$

Then $\mathfrak{K}$ is invertible and

$$\mathfrak{K} \simeq \mathcal{O}_B(1). \tag{9.1}$$

More precisely,

$$\det R_{g_1} = -\frac{1}{\tau}, \qquad \det R_{g_2} = \frac{1}{\tau}, \qquad \det R_{g_0} = 1, \tag{9.2}$$

and at the fixed points

$$\det R_{g_j}(z_j) = \zeta_j^{a_j}, \qquad (a_1, a_2) = (2, 1). \tag{9.3}$$

The form e generates $\mathfrak{K}$ at the cusp, while a unit times $s_j^{-a_j}e$ generates it at p_j.

Proof. Equation (2.3d) in theorem 6.1 gives

$$R_g(z)\Pi(z)\lambda = \Pi(gz)A_g\lambda.$$

Using

$$A_1\widehat{w} = \widehat{u} - \widehat{w}, \quad A_1\widehat{\delta} = \widehat{\delta}, \qquad A_2\widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2\widehat{\delta} = \widehat{\delta},$$

and writing $\tau^\sharp = \tau(g_j z)$, $\mu^\sharp = \mu(g_j z)$, direct multiplication gives

$$R_{g_1} = \begin{pmatrix} \tau^\sharp - 1 & 0 \\ \mu^\sharp & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} -\tau^\sharp & 0 \\ 1 - \mu^\sharp & 1 \end{pmatrix}.$$

The transformation laws $\tau \circ g_1 = (\tau - 1)/\tau$ and $\tau \circ g_2 = -1/\tau$ prove (9.2) at g_1, g_2; the cusp formulas give $R_{g_0} = I$. Since $\widehat{\delta}$ is fixed, one eigenvalue is 1. At $\tau(z_1) = \rho = e^{\pi i/3}$, the other is $\rho - 1 = e^{2\pi i/3} = \zeta_1^2$; at $\tau(z_2) = i$, it is $-i = \zeta_2$. This proves (9.3).

We determine the extension over a finite filling without suppressing the ramification. On J_j', the nowhere-zero form $ds_j \wedge e$ transforms by

$$(\widetilde{g}_j^{\log})^*(ds_j \wedge e) = \chi_j(s_j)\, ds_j \wedge e, \qquad \chi_j = \zeta_j \det R_{g_j}. \tag{9.4}$$

Iteration m_j times shows $\prod_{r=0}^{m_j-1} \chi_j(\zeta_j^r s) = 1$. Write $\chi_j(s) = \chi_j(0)e^{\psi(s)}$, with $\psi(0) = 0$. The product identity says that the power series of ψ has no terms whose degree is divisible by m_j. Hence

$$\varphi(s) - \varphi(\zeta_j s) = \psi(s)$$

has a holomorphic solution, obtained coefficientwise by division by $1 - \zeta_j^n$. Therefore

$$s^{k_j} e^{\varphi(s)} ds \wedge e, \qquad k_j = m_j - 1 - a_j, \tag{9.5}$$

is invariant: $\chi_j(0) = \zeta_j^{1+a_j}$, so invariance is equivalent to $k_j + 1 + a_j \equiv 0 \pmod{m_j}$. The finite action is free by theorem 6.5, so (9.5) descends to a local generator of K_X with divisor $k_j S_j$. Dividing by

$$f^* dt_j = m_j s_j^{m_j - 1} ds_j$$

shows that a unit times $s_j^{-a_j} e$ is the corresponding generator of $\omega_{X/B}$.

At the cusp, in every smooth toric chart the unimodular monomial change of coordinates turns

$$\Omega_0 = \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge dt_c$$

into a unit times the ordinary coordinate volume form. Thus it generates the canonical line. It is invariant under translations and under the unimodular shears defining the quotient. Under the punctured-overlap map, $\Omega_0 = (2\pi i)^2 dt_c \wedge e$, so e generates $\mathfrak{K}$ at p_0.

It remains to compute the degree. The modular-form calculation already used in the proof of theorem 6.1 gives

$$\Theta = \left(\frac{E_4^2 E_6^{1/2}}{\Delta_{\mathrm{mod}}} \right) \circ \tau.$$

The square root exists because the zeros of $E_6 \circ \tau$ have even order. The weights $8, 3, 12$ give the same automorphy factor as e; the sign at z_2 is fixed by its linearizing coordinate. The orders are

$$\operatorname{ord}_{z_1} \Theta = 2 = a_1, \qquad \operatorname{ord}_{z_2} \Theta = 1 = a_2, \qquad \Theta = t_c^{-1} \cdot (\text{unit}) \tag{9.6}$$

at the cusp. Consequently $\mathsf{e} = \Theta^{-1} e$ is single-valued. At p_j it is a unit times $s_j^{-a_j} e$; at p_0 it is a unit times $t_c e$. Hence it is a global section of $\mathfrak{K}$ with the single simple zero p_0. This proves $\mathfrak{K} \simeq \mathcal{O}_B(p_0) \simeq \mathcal{O}_B(1)$. Finally, $\mathfrak{K}$ is torsion free of generic rank one on the smooth curve B, hence is invertible. □

Lemma 9.2 (exact order of the finite-fibre normal line)**.** ***[VERIFIED]*** *For $j = 1, 2$,*

$$\mathcal{O}_X(S_j)|_{S_j} = N_{S_j/X}$$

has exact order m_j in $\mathrm{Pic}(S_j)$*.*

Proof. On the finite cover of theorem 6.5, the central torus $A_0^{(j)} = \{s_j = 0\}$ maps to S_j as a connected principal $G_j \simeq \mathbb{Z}/m_j$-cover. Its normal line is trivial, and a generator of G_j acts on the fibre by $s_j \mapsto \zeta_j s_j$. Thus $N_{S_j/X}$ is associated to this principal cover by the character ζ_j, and its m_j-th power is trivial.

Suppose its k-th power were trivial for some $0 < k < m_j$. Pulling a nowhere-zero trivializing section back to $A_0^{(j)}$ would give a nowhere-zero holomorphic function h satisfying

$$h(gx) = \zeta_j^{-k} h(x).$$

Every holomorphic function on the compact connected torus $A_0^{(j)}$ is constant. The displayed equation would force $\zeta_j^k = 1$, contrary to $0 < k < m_j$. Hence the order is exactly m_j. □

9.2 The cusp structure sheaf

Lemma 9.3 (structure-sheaf cohomology of the cusp fibre)**.** ***[VERIFIED]*** *The reduced normal-crossings fibre W satisfies*

$$h^q(W, \mathcal{O}_W) = (1, 2, 1) \qquad (q = 0, 1, 2). \tag{9.7}$$

The natural map

$$H^1(W; \mathbb{C}) \longrightarrow H^1(W, \mathcal{O}_W) \tag{9.8}$$

is an isomorphism. In degree two the corresponding map is surjective and is injective on the kernel of pullback to the normalization $\nu : \mathrm{dP}_6 \to W$. Finally,

$$\bigwedge^2 H^1(W, \mathcal{O}_W) \longrightarrow H^2(W, \mathcal{O}_W) \tag{9.9}$$

is an isomorphism.

Proof. Let $D_1, D_2, D_3 \simeq \mathbb{P}^1$ be the double curves and P, Q the two triple points, with the branch order fixed in the proof of theorem 7.1. The normalization complex is

$$0 \longrightarrow \mathcal{O}_W \longrightarrow \nu_* \mathcal{O}_{\mathrm{dP}_6} \xrightarrow{\delta_0} \bigoplus_{k=1}^{3} \mathcal{O}_{D_k} \xrightarrow{\delta_1} \mathbb{C}_P \oplus \mathbb{C}_Q \longrightarrow 0, \tag{9.10}$$

where δ_0 is the difference of the two branch restrictions and

$$\delta_1(g_1, g_2, g_3) = (g_1(P) - g_2(P) + g_3(P), g_1(Q) - g_2(Q) + g_3(Q)).$$

For completeness, exactness is stalkwise. At a generic double point it is the fibre-product sequence

$$0 \to \mathbb{C}\{x, y, z\}/(xy) \to \mathbb{C}\{y, z\} \oplus \mathbb{C}\{x, z\} \to \mathbb{C}\{z\} \to 0.$$

At a triple point it is the augmented branch complex for $\mathbb{C}\{x, y, z\}/(xyz)$: three branch rings, three pairwise intersection rings, and the common residue field, with alternating differences. The augmented two-simplex complex is exact; successively lifting compatible power series proves exactness of the analytic rings. Away from the singular locus the first arrow is an isomorphism.

The same formulas give an exact complex with $\mathbb{C}$ in place of $\mathcal{O}$. Since dP_6 is rational and each D_k is a rational curve, the positive structure-sheaf cohomology of every term on the right of (9.10) vanishes. Hypercohomology therefore reduces to

$$\mathbb{C} \xrightarrow{0} \mathbb{C}^3 \xrightarrow{d_1} \mathbb{C}^2, \qquad d_1(a_1, a_2, a_3) = (s, s), \quad s = a_1 - a_2 + a_3. \tag{9.11}$$

Its cohomology has dimensions $1, 2, 1$, proving (9.7).

For the constant-sheaf complex, the row of degree zero is again (9.11), the row of degree one vanishes, and the remaining degree-two row begins with $H^2(\mathrm{dP}_6; \mathbb{C})$. The comparison induced by $\mathbb{C}_W \hookrightarrow \mathcal{O}_W$ is the identity on (9.11) and zero on the positive rows of the target. It follows that (9.8) is an isomorphism, while in degree two it maps the dual-complex summand $\ker \nu^*$ isomorphically onto $H^2(W, \mathcal{O}_W)$.

Choose the basis $\gamma_W, u_W \in H^1(W; \mathbb{Z})$ whose intrinsic specializations are γ, u, respectively; it exists and is unique by theorem 7.1. Their cup product is nonzero, because its specialization is $\gamma \wedge u \neq 0 \in H^2(F; \mathbb{Z})$. Its pullback to dP_6 is zero since $H^1(\mathrm{dP}_6; \mathbb{C}) = 0$. The degree-two comparison therefore sends it to a nonzero element of $H^2(W, \mathcal{O}_W)$. The images of γ_W, u_W form a basis by (9.8), and both sides of (9.9) are one-dimensional. Hence (9.9) is an isomorphism. □

9.3 Direct images, base change, and the Picard line

Theorem 9.4 (structure-sheaf direct images)**.** ***[VERIFIED]*** *For the constructed map $f : X \to B = \mathbb{P}^1$,*

$$f_*\mathcal{O}_X = \mathcal{O}_B, \qquad R^1 f_*\mathcal{O}_X \simeq \mathcal{O}_B \oplus \mathcal{O}_B(-1), \qquad R^2 f_*\mathcal{O}_X \simeq \mathcal{O}_B(-1). \tag{9.12}$$

All three direct images are locally free and every base-change morphism

$$(R^q f_*\mathcal{O}_X) \otimes k(b) \longrightarrow H^q(X_b, \mathcal{O}_{X_b}), \qquad 0 \le q \le 2, \tag{9.13}$$

is an isomorphism, including at the two nonreduced fibres. Moreover the invariant class γ defines the trivial summand in $R^1 f_*\mathcal{O}_X$ and is nowhere zero on B.

Proof. The proper map has connected fibres and B is normal, so Stein factorization gives $f_*\mathcal{O}_X = \mathcal{O}_B$.

The map is flat. A local parameter on B pulls back to a nonzero element of the regular local domain $\mathcal{O}_{X,x}$; if it vanished identically near one point, analytic continuation on the connected manifold would make f constant. Thus $\mathcal{O}_{X,x}$ is torsion-free, hence flat, over the discrete valuation ring $\mathcal{O}_{B,f(x)}$.

Over B°, the fibres are two-dimensional complex tori and their structure-sheaf cohomology dimensions are $1, 2, 1$. At the cusp the same dimensions hold by theorem 9.3. Grauert's theorem therefore gives local freeness of ranks $1, 2, 1$ and base change on a neighbourhood of $B^\circ \cup \{p_0\}$; the precise analytic statement is [6, Chapter III, §4, Theorem 4.12], originally due to Grauert [7].

At p_j, use the finite cover $J_j' \to N_j$. If $\widetilde{f} : J_j' \to \Delta_j$ is the smooth torus family and $\pi(s) = s^{m_j}$, exactness of finite-group invariants in characteristic zero gives

$$R^q f_*\mathcal{O}_{N_j} = \left(\pi_* R^q \widetilde{f}_*\mathcal{O}_{J_j'}\right)^{G_j}. \tag{9.14}$$

After averaging, a free eigenframe v may be chosen for each character ν. The invariant sections are exactly $s^r v\, \mathcal{O}_{D_j}$, where $0 \le r < m_j$ is determined by $\nu\zeta_j^r = 1$. Hence (9.14) is free of ranks $1, 2, 1$. Higher direct images vanish by the fibre-dimension bound, so these are all the nonzero direct images. Since all direct images are locally free, the analytic cohomology-and-base-change theorem, applied by descending induction from $q = 2$, proves (9.13) at the finite fibres as well. In the notation of [6, Chapter III, §3, Theorem 3.4, Corollaries 3.5, 3.7, and 3.10]. Corollary 3.10 says that, for each fixed q, freeness of $R^p f_*\mathcal{O}_X$ for every $p \ge q$ is equivalent to cohomological flatness in all dimensions $p \ge q$. Applying this successively with $q = 2, 1, 0$ gives the displayed base-change maps in every relevant degree. This is a local statement on the base and so applies equally to the nonreduced scheme-theoretic fibres.

Write $G = R^1 f_* \mathcal{O}_X$. We construct its distinguished section without using a cellular collapse. On B°, the monodromy-invariant integral class γ maps to a holomorphic section $\gamma^{0,1}$ of G. It is nowhere zero there: if its image in $H^{0,1}(F)$ vanished, then $\gamma \in H^{1,0}(F) \cap H^1(F;\mathbb{R}) = 0$. On a finite cover it is a constant invariant class and remains nonzero at $s_j = 0$, so it descends across p_j.

At the cusp, let $j : \Delta^* \hookrightarrow \Delta$. The natural specialization morphism

$$R^1 f_* \mathbb{C} \longrightarrow j_* R^1 f_* \mathbb{C}|_{\Delta^*}$$

is an isomorphism: away from 0 this is tautological, while at 0 it is exactly the isomorphism from $H^1(W;\mathbb{C})$ onto the T_0-invariant subspace proved in theorem 7.1. Hence the invariant classes γ, u extend as sections of $R^1 f_* \mathbb{C}$ on the whole disc. The natural morphism induced by $\mathbb{C} \hookrightarrow \mathcal{O}$,

$$R^1 f_* \mathbb{C} \otimes \mathcal{O}_\Delta \longrightarrow G|_\Delta,$$

sends them at the central point to the two classes in $H^1(W, \mathcal{O}_W)$. These form a basis by theorem 9.3. Thus $\gamma^{0,1}$ extends across the cusp and is nowhere zero everywhere. In particular $\mathcal{O}_B \gamma^{0,1} \subset G$ is a line subbundle.

Relative duality applies to the proper flat Gorenstein map f: X is smooth, the base is a smooth curve, and $\omega_{X/B}$ is invertible. Using theorem 9.1, it gives

$$R^2 f_* \mathcal{O}_X \simeq \mathfrak{K}^\vee \simeq \mathcal{O}_B(-1). \tag{9.15}$$

The relative theorem is the one of Ramis–Ruget–Verdier [5, pp. 261–283]; the absolute singular dualizing-complex comparison is independently recorded in [4, Theorems 1–2 and Proposition 2].

The relative cup product

$$\bigwedge^2 G \longrightarrow R^2 f_* \mathcal{O}_X \tag{9.16}$$

is an isomorphism. It is so on every smooth torus fibre. At the cusp it is so by (9.13) and theorem 9.3. At p_j, choose on the cover an eigenframe v_1, v_2, with $v_1 = \gamma^{0,1}$ invariant. Its invariant exponent is $r_1 = 0$. The action on $v_1 \wedge v_2$ is dual to the Hodge character in (9.3), so its invariant generator is $s_j^{a_j}(v_1 \wedge v_2)$. The invariant generator for the second line is consequently $s_j^{a_j} v_2$, because $0 < a_j < m_j$. Wedge product sends $v_1 \wedge (s_j^{a_j} v_2)$ to exactly the invariant generator of the right side of (9.16), with no wrap modulo m_j.

Consequently $\det G \simeq R^2 f_* \mathcal{O}_X \simeq \mathcal{O}_B(-1)$. The quotient of G by its trivial subbundle is $\mathcal{O}_B(-1)$, and the extension splits because

$$\operatorname{Ext}^1(\mathcal{O}_B(-1), \mathcal{O}_B) = H^1(B, \mathcal{O}_B(1)) = 0.$$

This proves (9.12) and identifies the trivial summand. □

Corollary 9.5 (the Picard parameter and fibrewise restriction)**.** ***[VERIFIED]*** *One has*

$$H^1(X, \mathcal{O}_X) = \mathbb{C}, \qquad \operatorname{Pic}(X) = \operatorname{Pic}^0(X) \simeq (\mathbb{C}, +). \tag{9.17}$$

For every scheme-theoretic fibre X_b, restriction induces an injective complex-linear map

$$\iota_b : H^1(X, \mathcal{O}_X) \longrightarrow H^1(X_b, \mathcal{O}_{X_b}). \tag{9.18}$$

On a smooth fibre its image is the complex line generated by the $(0,1)$-image of γ.

Proof. The Leray spectral sequence and $H^1(B, \mathcal{O}_B) = 0$ identify

$$H^1(X, \mathcal{O}_X) = H^0(B, R^1 f_* \mathcal{O}_X) = \mathbb{C}$$

by theorem 9.4. Since the verified integral topology gives $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, the analytic exponential sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{u \mapsto e^{2\pi i u}} \mathcal{O}_X^* \to 1$$

identifies $H^1(X, \mathcal{O}_X)$ with $\mathrm{Pic}(X)$ and shows that every line bundle is topologically trivial. Under Leray, restriction is evaluation of a global section of $R^1 f_* \mathcal{O}_X$, followed by base change. The only global section is a scalar multiple of the nowhere-zero section $\gamma^{0,1}$, so evaluation is injective at every point and has the stated image on a smooth fibre. □

9.4 One line bundle non-torsion on every fibre

Theorem 9.6 (uniform non-torsion)**.** ***[VERIFIED]*** *There is a line bundle $L \in \mathrm{Pic}(X)$ such that $L|_{X_b}$ is non-torsion for every scheme-theoretic fibre $X_b = f^*(b)$, and $L|_{S_j}$ is non-torsion for $j = 1, 2$. More precisely, identify $\mathrm{Pic}(X)$ with $\mathbb{C}$, $z \mapsto L_z$. Every z outside the union of $\mathbb{R}$ with a countable set has these properties.*

Proof. The pullback $f^*\mathcal{O}_B(1)$ is nontrivial: by the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_B$,

$$h^0(X, f^*\mathcal{O}_B(1)) = h^0(B, \mathcal{O}_B(1)) = 2,$$

whereas $h^0(X, \mathcal{O}_X) = 1$. Rescale the additive coordinate in theorem 9.5 so that

$$L_{12} = f^*\mathcal{O}_B(1). \tag{9.19}$$

Fix a smooth fibre F_q. Put

$$\Lambda_q = \iota_q^{-1}\Big(\mathrm{im}[H^1(F_q;\mathbb{Z}) \to H^1(F_q, \mathcal{O}_{F_q})]\Big) \subset \mathbb{C}. \tag{9.20}$$

The integral image is a lattice in the underlying real vector space of $H^1(F_q, \mathcal{O}_{F_q})$, so Λ_q is discrete. It contains 12, because (9.19) is trivial on every fibre. The exponential sequence on F_q shows

$$L_z|_{F_q} \text{ is torsion} \quad \Longleftrightarrow \quad z \in \mathbb{Q}\Lambda_q. \tag{9.21}$$

If $\mathrm{rk}\,\Lambda_q = 1$, its nonzero element 12 forces $\Lambda_q \subset \mathbb{R}$; all its bad parameters are real.

Suppose $\mathrm{rk}\,\Lambda_q = 2$. By theorem 9.5, the image of ι_q is the line generated by the image of γ. Thus there is an integral class

$$\lambda = a\gamma + bu + cw + d\delta \notin \mathbb{Q}\gamma \tag{9.22}$$

whose $(0,1)$-image lies on that line. On the covering family, the four marked lattice columns are

$$\pi_{\widehat{\gamma}} = (6\mu, \beta), \quad \pi_{\widehat{u}} = (\tau, \mu), \quad \pi_{\widehat{w}} = (1, 0), \quad \pi_{\widehat{\delta}} = (0, 1). \tag{9.23}$$

Every real cohomology class has a unique expression

$$2\,\mathrm{Re}(\bar{\xi}\, d\zeta_1 + \bar{\upsilon}\, d\zeta_2).$$

Evaluating it on (9.23) gives

$$\begin{aligned} 2\,\mathrm{Re}\,\bar{\xi} &= c, & 2\,\mathrm{Re}\,\bar{\upsilon} &= d, \\ 2\,\mathrm{Re}(\bar{\xi}\tau + \bar{\upsilon}\mu) &= b, & 2\,\mathrm{Re}(6\bar{\xi}\mu + \bar{\upsilon}\beta) &= a. \end{aligned} \tag{9.24}$$

For γ, the first two right sides vanish. The third equation then shows that its $(0,1)$-line is represented, up to a nonzero scalar, by $(\mathrm{Im}\,\mu, -\mathrm{Im}\,\tau)$; the fourth equation only fixes the scalar. Write $\bar{\xi} = c/2 + ir$, $\bar{\upsilon} = d/2 + is$. The condition that the class in (9.22) have $(0,1)$-image on the γ-line is

$$\bar{\xi}\,\mathrm{Im}\,\tau + \bar{\upsilon}\,\mathrm{Im}\,\mu = 0.$$

Its real part, together with the third equation of (9.24), is equivalent to the single holomorphic equation

$$c\tau + d\mu - b = 0. \tag{9.25}$$

Neither a nor β occurs, as adding a multiple of γ must not change the condition.

The functions $1, \tau, \mu$ are linearly independent. Applying the cusp transformation $(\tau, \mu) \mapsto (\tau - 1, \mu)$ to a constant linear relation first kills the coefficient of τ. If the remaining relation made μ constant, the order-four law $\mu \circ g_2 = 1 + \mu/\tau$, with nonconstant τ, would force an impossible identity. Hence all coefficients vanish.

For every integral $\lambda \notin \mathbb{Q}\gamma$, equation (9.25) is therefore a nonzero holomorphic function on the covering Riemann surface. Its zero set is discrete and countable. There are countably many λ, so their union, and its image in B°, are countable. This image contains every q for which $\operatorname{rk} \Lambda_q = 2$. Each such fibre contributes the countable bad set $\mathbb{Q}\Lambda_q$; all rank-one fibres together contribute only parameters in $\mathbb{R}$. Thus the smooth fibres exclude only $\mathbb{R}$ together with a countable set.

There remain the three scheme-theoretic special fibres. By theorems 9.4 and 9.5, restriction

$$\mathbb{C} \longrightarrow H^1(X_b, \mathcal{O}_{X_b})$$

is complex-linear and injective for each of them. The exponential sequence and countability of $H^1(X_b; \mathbb{Z})$ show that the parameters for which $L_z|_{X_b}$ is torsion form a countable set.

Finally consider the reductions S_j. The restriction $\varphi_j : \mathbb{C} \to H^1(S_j, \mathcal{O}_{S_j})$ is complex-linear. It is nonzero. Write $\mathcal{O}_X(S_j) = L_{z_j}$. Since

$$\mathcal{O}_X(m_j S_j) = f^* \mathcal{O}_B(1) = L_{12},$$

the additive parameter gives $z_j = 12/m_j$, while theorem 9.2 says that

$$L_{12/m_j}|_{S_j} = \mathcal{O}_X(S_j)|_{S_j}$$

is nontrivial. If φ_j were zero, the exponential sequence and the vanishing of the restricted first Chern class would make every $L_z|_{S_j}$ trivial, a contradiction. Thus φ_j is injective, and the same exponential-sequence argument makes its torsion parameters countable. Removing these two additional countable sets proves the theorem. □

Remark 9.7 (provenance of the repair). The period equation $c\tau + d\mu - b = 0$ and the countability strategy are the candidate's. The exact normal-character proof above makes the order of $\mathcal{O}_X(S_j)|_{S_j}$ explicit, while the cusp extension in theorem 9.4 replaces the candidate's collapse-dependent argument by the independently verified intrinsic nearby-cycle map. Identifying the restriction line with the global $\gamma^{0,1}$-summand also removes the candidate's separate continuity argument for a moving integral lift.

10 The alternate collapse and Mayer–Vietoris route

The nearby-cycle/Leray proof in section 7 gives a verified independent route to the topology of X. This section keeps the manuscript's independent cellular-collapse/Mayer–Vietoris route under audit. The integer algebra in theorems 5.5 and 5.6 is complete, but the geometric realization of the printed homology matrices is not yet needed or fully validated. Write the relevant decomposition as $X = U \cup N$, with $N = \bigcup_j N_j$ and $U \cap N$ the disjoint union of oriented collars. The Mayer–Vietoris map is

$$\delta_q = (i_{U*}, -i_{N*}) : H_q(U \cap N; \mathbb{Z}) \longrightarrow H_q(U; \mathbb{Z}) \oplus H_q(N; \mathbb{Z}). \tag{4.1}$$

The assertion requiring geometric proof is not the Smith form of a printed matrix but

$$\alpha_q = [\delta_q]_{\mathcal{B}_q} \tag{4.2}$$

for bases $\mathcal{B}_q$ constructed from actual primitive oriented cycles.

10.1 A complete abstract cusp-retraction lemma

Proposition 10.1 (equivariant topological spine)**.** ***[VERIFIED]*** *Let A be a contractible manifold with boundary and $A_0 \subset \partial A$ a contractible union of boundary components. Let a group Γ act freely and properly discontinuously on (A, A_0). Assume $Q = A/\Gamma$ and $H = A_0/\Gamma$ are manifolds, Q is compact, $H \hookrightarrow Q$ is collared, and the two quotient maps are connected coverings with deck group Γ. If the inclusion identifies the two deck groups by the identity automorphism of Γ, then $H \hookrightarrow Q$ is a strong deformation retract. The retraction lifts uniquely from the identity lift to a Γ-equivariant strong deformation retraction of A onto A_0.*

Proof. Because A and A_0 are contractible universal covers, $\pi_n(Q) = \pi_n(H) = 0$ for $n \geq 2$, while $\pi_1(Q) \cong \Gamma \cong \pi_1(H)$. By hypothesis the inclusion induces the identity on Γ, so it is a weak homotopy equivalence. Manifolds have the homotopy type of CW complexes; Whitehead's theorem [9, Theorem 4.5] therefore makes the inclusion a homotopy equivalence.

The collar makes (Q, H) an NDR pair, hence the inclusion is a cofibration. We record the standard cofibration argument. Choose a homotopy inverse $r_0 : Q \to H$. Since $r_0|_H \simeq \mathrm{id}_H$, the homotopy extension property modifies r_0 through maps $Q \to H$ to a retraction r satisfying $r|_H = \mathrm{id}_H$. The maps id_Q and $i \circ r$ are homotopic and agree on H. Applying the homotopy extension property once more makes that homotopy stationary on H. It is therefore a strong deformation retraction.

Let $P : A \to Q$ be the covering. Lift the deformation homotopy from the initial lift id_A. Since the base homotopy fixes H and the initial lift fixes A_0, uniqueness of path lifting makes the lifted homotopy fix A_0. For $\gamma \in \Gamma$, both $\widetilde{H}_t \circ \gamma$ and $\gamma \circ \widetilde{H}_t$ lift the same map and agree at $t = 0$; uniqueness gives equality for every t. The lift is therefore Γ-equivariant and ends in A_0. □

This proves the formal retraction step once its hypotheses are established. The manifold, freeness, proper discontinuity, and quotient assertions for the infinite toric space are now proved in theorem 6.3; the candidate-specific collar and contractibility assertions used by this alternate route remain to be checked.

10.2 The remaining candidate-specific obligations

Proof obligation 10.2 (C1: finite collapse model)**.** **[OPEN]** The analytic quotient prerequisites are discharged by theorem 6.3. For the alternate collapse proof, construct the claimed finite quotient CW structure, verify its attaching maps on all common faces, and check the collar, contractibility, and identity deck group hypotheses needed by theorem 10.1.

Proof obligation 10.3 (C2–C3: collapse and specialization)**.** **[OPEN]** Construct the collapse on each toric stratum, prove compatibility on all face inclusions and $\bar{\Lambda}$-equivariance, and compute its map on integral cellular chains with oriented primitive bases. Then prove that this topological collapse map agrees with the nearby-cycle specialization

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z})$$

. The image, torsion, and saturation of specialization itself are proved in theorem 7.1 for the reconstructed cusp model. What remains for this alternate route is the model-specific equality of that map with the cellular collapse and the oriented chain matrix. SGA 7 supplies formal nearby cycles [22]; it does not supply that equality.

Proof obligation 10.4 (V1: based van Kampen inclusions)**.** **[VERIFIED]** The actual overlap coordinate maps, with the zero section as base locus and clockwise meridians, show that the finite fillings impose $\widehat{\rho}_j^{m_j} = t_{v_j}$ with the same sign, the cusp filling kills Λ_{tor} and its toric meridian, and the tautological gluing has zero translation drift. Thus the abstract terminal presentation is the actual $\pi_1(X)$. The deck generators, common clockwise convention, zero-section disc, normal closure, boundary inclusion $\Lambda \to \Lambda/\Lambda_{\mathrm{tor}}$, $\widetilde{a}_0 \mapsto 1$, and absence of extra whisker drift are proved in theorem 7.6.

Proof obligation 10.5 (MV1 and the remaining comparison)**.** **[OPEN]** Prove (4.2) for every degree used in the homology calculation. The integral Leray transgression and its candidate sign are already derived in theorem 7.3; the higher transgressions are theorem 7.4. What remains is a commuting comparison diagram between those maps and (4.1), listing every nearby-cycle, meridian, transfer, and basis identification shared by the two routes. Agreement of their terminal integers is not itself such a diagram.

Validation, repair, or disproof test 10.6 (immediate topological falsifiers)**.** The alternate collapse certificate fails if any actual map has a nonprimitive image where the printed matrix assumes a primitive one, if specialization contributes an omitted torsion class or nonunit index, if a meridian sign changes $p = -1$ to a nonunit, or if the geometric Leray transgression differs from $\pm p\omega$ by a nonunit factor.

10.3 Complete conditional recognition theorem

Lemma 10.7 (cohomological closure in dimension six)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected, oriented six-manifold. If*

$$\pi_1(X) = 0, \qquad H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \tag{4.3}$$

then $H_k(X;\mathbb{Z})$ is $\mathbb{Z}$ for $k = 0, 6$ and zero for $1 \le k \le 5$.

Proof. The Hurewicz map in degree one identifies $H_1(X;\mathbb{Z}) = \pi_1(X)^{\mathrm{ab}} = 0$. The universal coefficient sequence

$$0 \longrightarrow \mathrm{Ext}(H_1(X),\mathbb{Z}) \longrightarrow H^2(X;\mathbb{Z}) \longrightarrow \mathrm{Hom}(H_2(X),\mathbb{Z}) \longrightarrow 0$$

shows that $\mathrm{Hom}(H_2(X),\mathbb{Z}) = 0$, so the finitely generated group $H_2(X)$ is torsion. The next universal coefficient sequence contains an injection

$$\mathrm{Ext}(H_2(X),\mathbb{Z}) \hookrightarrow H^3(X;\mathbb{Z}) = 0.$$

For a finitely generated torsion group $A = \bigoplus_i \mathbb{Z}/n_i$, one has $\mathrm{Ext}(A,\mathbb{Z}) = \bigoplus_i \mathbb{Z}/n_i$; hence $H_2(X) = 0$. Poincaré duality gives

$$H_3(X) \simeq H^3(X) = 0, \quad H_4(X) \simeq H^2(X) = 0, \quad H_5(X) \simeq H^1(X) = 0.$$

Finally connectedness and orientation give $H_0(X) = H_6(X) = \mathbb{Z}$. □

Theorem 10.8 (sphere recognition from the exact inputs)**.** ***[CONDITIONAL]*** *Let X be a smooth, closed, connected six-manifold. If*

$$\pi_1(X) = 0, \qquad H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5, \end{cases} \tag{4.4}$$

then X is diffeomorphic to the standard S^6.

Proof. A closed smooth manifold has the homotopy type of a finite CW complex. We first prove successively that $\pi_i(X) = 0$ for $1 \le i \le 5$. The case $i = 1$ is assumed. Suppose $2 \le n \le 5$ and $\pi_i(X) = 0$ for $i < n$. Then X is $(n-1)$-connected, so the Hurewicz homomorphism

$$h_n : \pi_n(X) \longrightarrow H_n(X;\mathbb{Z})$$

is an isomorphism by the Hurewicz theorem [9, Theorem 4.32]. The target vanishes by (4.4), hence $\pi_n(X) = 0$. Induction proves the assertion.

Now X is 5-connected. Hurewicz in degree six gives $\pi_6(X) \cong H_6(X;\mathbb{Z}) \cong \mathbb{Z}$. Choose a map $g : S^6 \to X$ whose homotopy class maps to a chosen orientation generator. It has degree one. Hence g_*

is an isomorphism in degrees zero and six; in every other degree both integral homology groups vanish. Thus g is an integral homology equivalence. Both spaces are simply connected CW complexes, so the homological Whitehead theorem [9, Corollary 4.33] makes g a homotopy equivalence. Therefore X is a smooth homotopy 6-sphere.

Smale's generalized Poincaré theorem in dimensions greater than four [11, Theorem A] identifies the underlying topological manifold with S^6. Smooth homotopy 6-spheres up to oriented h-cobordism form the group Θ_6; the opening table of Kervaire–Milnor gives $|\Theta_6| = 1$ [12]. Hence there is no exotic smooth structure in dimension six, and X is diffeomorphic to the standard sphere. □

Corollary 10.9 (abstract application boundary)**.** ***[CONDITIONAL]*** *Suppose V1 identifies the actual based van Kampen quotient with $G(\ell_0, \ell_1, \ell_2)$ of theorem 5.5, and suppose an independently constructed MV1/L1 comparison proves*

$$H^2(X;\mathbb{Z}) \cong H^3(X;\mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z}, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2. \tag{4.5}$$

If $|p| = 1$, then $X \cong_{\mathrm{diff}} S^6$.

Proof. For $|p| = 1$, theorem 5.5 and the assumed geometric identification give $\pi_1(X) = 0$, while (4.5) gives $H^2(X) = H^3(X) = 0$. Theorem 10.7 proves (4.4), and theorem 10.8 gives the claimed diffeomorphism. □

The corollary is an abstract implication. For the actual candidate, the intrinsic nearby-cycle/Leray and based van Kampen route now proves the hypotheses used by theorem 7.7. C1–C3, MV1, and L1 remain only the independent collapse/Mayer–Vietoris certificate; the additional open work in this section concerns that alternate certificate.

11 The non-normal fibre and the CDP obstruction

Campana–Demailly–Peternell prove that a compact complex threefold with $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction has $c_3(X) \leq 0$ [2, Theorem 2.2, p. 680]. Their corollary gives $a(X) = 0$ for a compact complex threefold homeomorphic to S^6 [2, Corollary 2.3, p. 680]. The candidate asserts all those hypotheses together with $c_3(X) = 2$ and $a(X) = 1$. This section separates:

(i) validity of one printed inference on p. 692 of [2];

(ii) validity of the published theorem by any other argument;

(iii) validity of the candidate's proposed non-normal-fibre mechanism; and

(iv) truth of the candidate's upstream construction and topology.

11.1 The exact p. 692 inference

Let S be the reduction of a fibre in a smooth threefold $\mathcal{X}$, and let A denote the line-bundle twist after duality. The two relevant sequences are

$$0 \longrightarrow N^*_{S/\mathcal{X}} \otimes A \longrightarrow \Omega^1_{\mathcal{X}}|_S \otimes A \xrightarrow{\rho} \Omega^1_S \otimes A \longrightarrow 0 \tag{5.1}$$

and

$$0 \longrightarrow \mathcal{T}(\Omega^1_S) \otimes A \longrightarrow \Omega^1_S \otimes A \longrightarrow \widetilde{\Omega}^1_S \otimes A \longrightarrow 0, \tag{5.2}$$

where $\mathcal{T}$ denotes sections supported on a nowhere dense analytic set. The opening reduction of [2, §7, p. 692] treats the vanishing of the global sections of the left end of (5.1) and the right end of (5.2) as

sufficient for the vanishing of the middle term of (5.1). That inference leaves untreated a section whose image under ρ is a nonzero torsion section. The omitted sheaf can be identified exactly. Put

$$\mathscr{T}_S := \ker(\Omega^1_S \to j_*\Omega^1_{S_{\mathrm{reg}}}), \qquad \widetilde{\Omega}^1_S = \Omega^1_S/\mathscr{T}_S,$$

and

$$\mathscr{K}_S := \ker(\Omega^1_X|_S \longrightarrow \widetilde{\Omega}^1_S).$$

Proposition 11.1 (the missing cotangent filtration)**.** *Let S be a reduced effective Cartier divisor in a smooth complex threefold X. There are natural short exact sequences*

$$0 \longrightarrow N^*_{S/X} \longrightarrow \mathscr{K}_S \longrightarrow \mathscr{T}_S \longrightarrow 0,$$

$$0 \longrightarrow \mathscr{K}_S \longrightarrow \Omega^1_X|_S \longrightarrow \widetilde{\Omega}^1_S \longrightarrow 0.$$

Proof. The conormal sequence is right exact. It is also left exact here. Locally write $S = (g = 0)$ with g reduced. If $h\,dg|_S = 0$, then h vanishes at the generic smooth point of every irreducible component, because dg is nonzero there. The ring $\mathcal{O}_S$ is reduced, so the intersection of its minimal primes is zero; hence $h = 0$. Thus

$$0 \to N^*_{S/X} \to \Omega^1_X|_S \to \Omega^1_S \to 0$$

is exact. By definition, $\mathscr{K}_S$ is the inverse image of $\mathscr{T}_S$ under the last surjection. Restriction to that inverse image is onto $\mathscr{T}_S$ and has kernel $N^*_{S/X}$, proving the first sequence. Composing with the quotient by $\mathscr{T}_S$ proves the second. □

Proposition 11.2 (the exact global-section criterion)**.** *Let A be a line bundle on S. If*

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) = 0,$$

then

$$H^0(S, \Omega^1_X|_S \otimes A) = H^0(S, \mathscr{K}_S \otimes A).$$

If also $H^0(S, N^*_{S/X} \otimes A) = 0$, then this group is canonically

$$\ker\Big[H^0(S, \mathscr{T}_S \otimes A) \xrightarrow{\delta_A} H^1(S, N^*_{S/X} \otimes A)\Big].$$

Hence the two vanishings printed in the p. 692 reduction imply the desired ambient vanishing if and only if δ_A is injective.

Proof. Tensoring the second sequence of theorem 11.1 by the locally free sheaf A preserves exactness; its global-section sequence gives the first equality. Tensor the first sequence and take global sections. If the conormal H^0 vanishes, exactness identifies the middle group with the displayed kernel. No condition on the torsion-free quotient controls that kernel. □

11.2 Local algebra of a reduced normal-crossings fibre

Let $g : \mathcal{X} \to \Delta$ be a proper holomorphic map from a smooth complex threefold to a disc, and put $S = g^{-1}(0)$. Assume S is reduced and has normal crossings: at every point there are coordinates (x, y, z) in which

$$g = ux, \qquad g = uxy, \qquad \text{or} \qquad g = uxyz \tag{5.3}$$

for a unit u. Let $\eta : \widetilde{S} \to S$ be the normalization, $D = S_{\mathrm{sing,red}}$, and $\widetilde{D} = \eta^{-1}(D)_{\mathrm{red}}$.

Lemma 11.3 (the conormal map is injective)**.** ***[VERIFIED]*** *In the setting above, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, and its generator maps to $dg|_S$. Multiplication by $dg|_S$ is injective, so*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{h \mapsto h\,dg} \Omega^1_{\mathcal{X}}|_S \longrightarrow \Omega^1_S \longrightarrow 0 \tag{5.4}$$

is exact.

Proof. The fibre ideal is the principal ideal (g) and is reduced, so $(g)/(g^2) \simeq \mathcal{O}_S$. Multiplication by a unit does not affect injectivity, and the term $f\,du$ vanishes after restriction to $f = 0$. It therefore suffices to use $f = x, xy, xyz$.

For $f = xy$, if a lift $\widetilde{h}$ satisfies $\widetilde{h}(y\,dx + x\,dy) = 0$ modulo (xy), then $\widetilde{h}y, \widetilde{h}x \in (xy)$. Hence $\widetilde{h} \in (x) \cap (y) = (xy)$ in the regular local ring, so $h = 0$ on S. For $f = xyz$, the three coefficient conditions say $\widetilde{h}yz, \widetilde{h}xz, \widetilde{h}xy \in (xyz)$; hence $\widetilde{h} \in (x) \cap (y) \cap (z) = (xyz)$. The case $f = x$ is immediate. Thus the conormal map is injective in every local model, proving (5.4). □

Lemma 11.4 (normalization–conductor exact sequence)**.** ***[VERIFIED]*** *The sequence*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{\alpha} \eta_*\mathcal{O}_{\widetilde{S}} \oplus \mathcal{O}_D \xrightarrow{\beta} \eta_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0, \tag{5.5}$$

$$\alpha(h) = (\eta^* h, h|_D), \qquad \beta(\widetilde{h}, k) = \widetilde{h}|_{\widetilde{D}} - \eta^* k,$$

is exact.

Proof. The assertion is local. Away from D, η is an isomorphism. At a double point $S = \{xy = 0\}$, a member of $\ker \beta$ is

$$((h_1(y,z), h_2(x,z)), k(z)), \qquad h_1(0,z) = h_2(0,z) = k(z).$$

The function

$$h = h_1(y,z) + h_2(x,z) - h_1(0,z)$$

on $\{xy = 0\}$ restricts to h_1, h_2 on the two branches and to k on their intersection. This proves middle exactness in the double model.

At a triple point $S = \{xyz = 0\}$, write the branch functions as $h_1(y,z)$, $h_2(x,z)$, $h_3(x,y)$. Membership in $\ker \beta$ is exactly

$$h_1(0,z) = h_2(0,z), \quad h_1(y,0) = h_3(0,y), \quad h_2(x,0) = h_3(x,0).$$

Then

$$\begin{aligned} h = {} & h_1(y,z) + h_2(x,z) + h_3(x,y) \\ & - h_1(0,z) - h_1(y,0) - h_2(x,0) + h_1(0,0) \end{aligned}$$

restricts to the prescribed function on every branch. Injectivity of α follows because S is reduced and η is an isomorphism over a dense open set. Surjectivity of β follows locally by lifting a function on the branch conductor to the smooth branch and taking $k = 0$. This proves exactness in every local model. □

Theorem 11.5 (conductor-supported ambient differential)**.** ***[VERIFIED]*** *In the setting (5.3), let A be a line bundle on S with $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$, and choose a nowhere-vanishing section e of $\eta^* A$. There is a unique nonzero*

$$s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A) \tag{5.6}$$

such that

$$\eta^* s = \eta^*(dg|_S) \otimes e, \qquad s|_D = 0. \tag{5.7}$$

Its image $\rho(s)$ in $\Omega^1_S \otimes A$ is supported on D and therefore torsion; its image in $\widetilde{\Omega}^1_S \otimes A$ is zero. Moreover:

(a) $\rho(s) = 0$ *if and only if* $e = \eta^*\sigma$ *for a nowhere-vanishing* $\sigma \in H^0(S, A)$*;*

(b) if S *is connected and* $A \not\simeq \mathcal{O}_S$*, then* $H^0(S, A) = 0$ *and* s *does not come from the conormal term in (5.1).*

Proof. In the double model, $d(xy) = y\,dx + x\,dy$ has coefficients in the reduced ideal (x, y) of D. In the triple model,

$$d(xyz) = yz\,dx + xz\,dy + xy\,dz$$

has coefficients in (xy, xz, yz), the reduced ideal of the union of the three double curves. Unit factors in (5.3) do not change this after restriction to S. Hence $\eta^*(dg|_S)$ restricts to zero on $\widetilde{D}$. Tensor (5.5) by the locally free sheaf $E = \Omega^1_{\mathcal{X}}|_S \otimes A$. The compatible pair

$$(\eta^*(dg|_S) \otimes e, 0)$$

therefore descends uniquely to s. On the smooth locus of S, $dg|_S$ is a nonzero conormal vector and e is nowhere zero, so $s \neq 0$.

On $S \setminus D$, the normalization is an isomorphism and $s = dg|_S \otimes e$. By (5.4) its intrinsic image vanishes there. Thus $\rho(s)$ is supported on D and lies in the torsion subsheaf; its image modulo torsion is zero.

If $\rho(s) = 0$, exactness of (5.4) after tensoring by A gives $s = dg|_S \otimes \sigma$ for a section σ of A. Pulling back and comparing with (5.7) on the dense open set where $dg \neq 0$ gives $\eta^*\sigma = e$ there, hence everywhere by holomorphic continuation. Since e never vanishes and η is surjective, σ never vanishes and trivializes A. The converse is immediate.

Finally suppose S is connected and $\sigma \in H^0(S, A)$. On each compact connected component of $\widetilde{S}$, write $\eta^*\sigma = c_i e$; the holomorphic function c_i is constant. If one c_i is zero and another is nonzero, connectedness of the finite union of irreducible components of S provides two such components meeting at a point. The value of σ at that point would be both zero and nonzero, a contradiction. Hence either every $c_i = 0$, so $\sigma = 0$, or every $c_i \neq 0$, so σ is nowhere zero and A is trivial. When A is nontrivial only the first case occurs. Since $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, the conormal global-section space is $H^0(S, A) = 0$, while $s \neq 0$. □

11.3 A proper counterexample to the printed inference

Theorem 11.6 (nodal-cubic test family)**.** ***[REFUTED]*** *The p. 692 implication described after (5.2) is false as a formal sheaf-theoretic statement, even for a proper family with smooth total space and an irreducible reduced normal-crossings central fibre.*

Proof. Let Δ be a sufficiently small disc with coordinate t and put

$$\mathcal{C} = \{([x : y : z], t) \in \mathbb{P}^2 \times \Delta : y^2 z = x^2(x + z) + tz^3\}. \tag{5.8}$$

This is proper over Δ. Its total space is smooth. On $z \neq 0$, the partial derivative of the defining equation with respect to t is $-z^3 \neq 0$. On $z = 0$, the equation forces $x = 0$; then $y \neq 0$ and the z-derivative is $y^2 \neq 0$. The central fibre C_0 is the irreducible nodal cubic $y^2 z = x^2(x + z)$, with node $[0 : 0 : 1]$.

Set

$$\mathcal{X} = \mathcal{C} \times \mathbb{P}^1, \qquad S = C_0 \times \mathbb{P}^1,$$

and let $g : \mathcal{X} \to \Delta$ be projection. Then $\mathcal{X}$ is a smooth threefold, g is proper, and S is an irreducible reduced normal-crossings surface. Its normalization is $\widetilde{S} \simeq \mathbb{P}^1 \times \mathbb{P}^1$.

Let p, q be the two inverse images of the node in the normalization of C_0. Fix $\lambda \in \mathbb{C}^* \setminus \{1\}$. Glue the fibres of the trivial line bundle on the normalization at p and q by the scalar λ. This defines a line bundle $A_\lambda \in \mathrm{Pic}^0(C_0)$ whose pullback is trivial. It is not trivial: a global section pulls back to a constant c satisfying $c = \lambda c$, so

$$H^0(C_0, A_\lambda) = 0. \tag{5.9}$$

Put $A = \mathrm{pr}_1^* A_\lambda$ on S. Then $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ and $H^0(S, A) = 0$. Because S is the full fibre, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, so the conormal-twist global sections vanish.

The natural map $\widetilde{\Omega}^1_S \to \eta_* \Omega^1_{\widetilde{S}}$ is injective: its kernel is supported where normalization is not an isomorphism, while the source is torsion-free by definition. Projection formula and $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ therefore give

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) \hookrightarrow H^0(\widetilde{S}, \Omega^1_{\widetilde{S}}) = 0, \tag{5.10}$$

the last equality following from $\widetilde{S} = \mathbb{P}^1 \times \mathbb{P}^1$. Nevertheless theorem 11.5 produces

$$0 \neq s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A).$$

Thus both vanishings invoked in the printed reduction hold while the desired ambient vanishing fails. The family (5.8) does not satisfy the global hypotheses of CDP Theorem 2.2, so this is a counterexample to the inference, not to the theorem. □

11.4 Cartier-divisor duality without a normality hypothesis

Theorem 11.7 (complete Cartier-fibre duality)**.** ***[VERIFIED]*** *Let X be a compact smooth complex threefold, $i : W \hookrightarrow X$ an effective Cartier divisor, and E a holomorphic vector bundle. Put $F = E|_W$ and*

$$\omega_W = (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

For $q = 0, 1, 2$ there are natural perfect pairings

$$H^q(W, F)^\vee \simeq H^{2-q}(W, F^\vee \otimes \omega_W). \tag{5.11}$$

Moreover $H^q(W, F) = 0$ for $q > 2$. No normality or reducedness hypothesis on W is used.

Proof. The Cartier divisor gives two short exact sequences

$$0 \to E(-W) \to E \to i_* F \to 0 \tag{5.12}$$

and, by adjunction,

$$0 \to E^\vee \otimes \omega_X \to E^\vee \otimes \omega_X(W) \to i_*(F^\vee \otimes \omega_W) \to 0. \tag{5.13}$$

Smooth Serre duality on the compact threefold gives functorial isomorphisms

$$H^r(X, G)^\vee \simeq H^{3-r}(X, G^\vee \otimes \omega_X) \tag{5.14}$$

for every vector bundle G [8, Theorem 9.37]. Naturality of the trace pairing says that the dual of the map on cohomology induced by $E(-W) \to E$ is the map induced by $E^\vee \otimes \omega_X \to E^\vee \otimes \omega_X(W)$.

Fix $q \in \{0, 1, 2\}$. Dualize the relevant five terms of the long exact sequence of (5.12) and use (5.14). They become

$$\begin{aligned} H^{2-q}(X, E^\vee \otimes \omega_X) &\longrightarrow H^{2-q}(X, E^\vee \otimes \omega_X(W)) \\ &\longrightarrow H^q(W, F)^\vee \\ &\longrightarrow H^{3-q}(X, E^\vee \otimes \omega_X) \longrightarrow H^{3-q}(X, E^\vee \otimes \omega_X(W)). \end{aligned} \tag{5.15}$$

The long exact sequence of (5.13) is the same five-term sequence with its middle group replaced by $H^{2-q}(W, F^\vee \otimes \omega_W)$. Naturality supplies a commutative diagram whose four outer vertical arrows are identities. The five lemma (or exactness at the three middle terms) gives the unique middle isomorphism (5.11). Finite dimensionality makes the pairing perfect.

For $q = 3$, the map $H^3(X, E(-W)) \to H^3(X, E)$ is dual under (5.14) to the injective map

$$H^0(X, E^\vee \otimes \omega_X) \hookrightarrow H^0(X, E^\vee \otimes \omega_X(W)).$$

It is therefore surjective. The long exact sequence of (5.12) gives $H^3(W, F) = 0$. For $q > 3$ both adjacent ambient cohomology groups vanish by the dimension-three Dolbeault resolution, so (5.12) gives $H^q(W, F) = 0$. This also agrees with analytic duality for the Gorenstein hypersurface W in Ramis–Ruget [4, Theorems 1–2 and Proposition 2]. □

11.5 Flatness and top-degree analytic base change

Lemma 11.8 (a nonconstant map from a manifold to a curve is flat)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a nonconstant holomorphic map from a connected complex manifold to a Riemann surface. Then f is flat. Every holomorphic vector bundle on X is f-flat.*

Proof. Fix $x \in X$, $c = f(x)$, and put $R = \mathcal{O}_{B,c}$, $S = \mathcal{O}_{X,x}$. The ring R is a discrete valuation ring and S is a regular local domain. The homomorphism $R \to S$ is injective. Indeed, if a nonzero germ $h \in R$ mapped to zero, $h \circ f$ would vanish on a neighbourhood of x. The zero set of a nonzero one-variable holomorphic germ is isolated, so f would be constant on that neighbourhood and therefore, by the identity theorem on connected X, constant everywhere.

Thus every nonzero element of R acts on the domain S as a nonzero element and hence as a nonzerodivisor. Therefore S is torsion-free as an R-module. A module over a discrete valuation ring is flat if and only if it is torsion-free [26, Tag 0AUW]; hence S is R-flat. This holds at every point, so f is flat. A vector bundle is locally a finite free $\mathcal{O}_X$-module, and a finite direct sum of R-flat modules is flat; hence it is f-flat. □

Proposition 11.9 (the exact analytic base-change input)**.** ***[VERIFIED]*** *Let $f : X \to B$ be proper between complex spaces, let $\mathcal{F}$ be coherent and f-flat, and fix $c \in B$. If*

$$H^3(X_c, \mathcal{F}|_{X_c}) = 0, \tag{5.16}$$

then the canonical fibre map

$$\beta_c^2 : (R^2 f_* \mathcal{F})_c \otimes_{\mathcal{O}_{B,c}} \mathbb{C}(c) \longrightarrow H^2(X_c, \mathcal{F}|_{X_c}) \tag{5.17}$$

is an isomorphism.

Proof. Apply Bănică–Stănăşilă, Chapter III, §3, Corollary 3.5, with $q = 3$ [6, pp. 120–121]. Its hypotheses are precisely properness, coherence, and flatness. Condition (5.16) first forces the degree-three direct image to vanish near c. The corollary then applies the equivalent base-change conditions of its preceding Theorem 3.4 in degree $q - 1 = 2$ and states that the canonical map (5.17) is bijective. This is the top-degree argument used in [3, Lemma 1.5]; it does not require the stronger, false assertion that all direct images are locally free. □

11.6 Conditional theorem for the candidate fibre

Theorem 11.10 (the candidate conductor mechanism, with all hypotheses)**.** ***[CONDITIONAL]*** *Assume:*

(H1) X is a connected compact smooth complex threefold and $f : X \to B$ is a nonconstant holomorphic map to a smooth compact curve;

(H2) $W = f^{-1}(p_0)$ is the full scheme-theoretic fibre, is reduced, and has the normal-crossings local models (5.3);

(H3) the normalization $\eta : \widetilde{W} \to W$ is the degree-six del Pezzo surface dP_6*;*

(H4) $H^2(X; \mathbb{Z}) = 0$.

Then, for every $L \in \mathrm{Pic}(X)$,

$$H^2(W, (T_X \otimes L)|_W) \neq 0 \quad \text{and} \quad R^2 f_*(T_X \otimes L) \neq 0. \tag{5.18}$$

Proof. Fix L and put

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Hypothesis (H4) gives $c_1(L^\vee \otimes \omega_X) = 0$. Pullback therefore gives $c_1(\eta^* A_L) = 0$. The surface dP_6 is the blow-up of $\mathbb{P}^2$ at three noncollinear points; the standard blow-up calculation gives $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$ [13, Chapter V]. In the exponential sequence, the kernel of

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

is the image of $H^1(\mathrm{dP}_6, \mathcal{O})$ and is therefore zero. Consequently

$$\eta^* A_L \simeq \mathcal{O}_{\widetilde{W}}. \tag{5.19}$$

Theorem 11.5 now gives

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0. \tag{5.20}$$

Because $W = f^*(p_0)$, one has $\mathcal{O}_X(W) = f^*\mathcal{O}_B(p_0)$, whose restriction to the fibre is trivial. Adjunction therefore gives $\omega_W \simeq \omega_X|_W$. Apply theorem 11.7 to $E = \mathcal{F}_L$:

$$\begin{aligned} H^2(W, \mathcal{F}_L|_W)^\vee &\simeq H^0(W, (\mathcal{F}_L|_W)^\vee \otimes \omega_W) \\ &= H^0(W, \Omega^1_X|_W \otimes A_L). \end{aligned}$$

The right side is nonzero by (5.20), proving the first assertion of (5.18). The same duality theorem gives $H^3(W, \mathcal{F}_L|_W) = 0$.

Since X is compact, f is proper. By theorem 11.8, f is flat and the vector bundle $\mathcal{F}_L$ is f-flat. Thus every hypothesis of theorem 11.9 is satisfied at p_0, and

$$(R^2 f_* \mathcal{F}_L)_{p_0} \otimes \mathbb{C}(p_0) \simeq H^2(W, \mathcal{F}_L|_W) \neq 0.$$

The stalk, and hence the coherent direct-image sheaf, is nonzero. This is the second assertion of (5.18). □

Corollary 11.11 (application to the constructed threefold)**.** ***[VERIFIED]*** *For the global construction of theorem 6.7, the cusp hypotheses (H1)–(H3) and the integral-topology hypothesis (H4) hold for the same threefold. Hence for every $L \in \mathrm{Pic}(X)$,*

$$H^2(W, (T_X \otimes L)|_W) \neq 0, \qquad R^2 f_*(T_X \otimes L) \neq 0.$$

Proof. Hypotheses (H1)–(H3) are theorems 6.3 and 6.7, and (H4) is theorem 7.5. Apply theorem 11.10. □

11.7 A second printed issue: abelianization of split extensions

Lemma 11.12 (abelianization of a semidirect product)**.** ***[VERIFIED]*** *If $1 \to K \to G \to Q \to 1$ is split, then*

$$G^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q, \tag{5.21}$$

where the subscript denotes coinvariants for the conjugation action of Q.

Proof. Choose a splitting and write $G = K \rtimes Q$. Let $N \triangleleft K$ be generated by $[K, K]$ and by all elements $qkq^{-1}k^{-1}$ with $q \in Q$, $k \in K$. It is normal in G: $[K, K]$ makes it stable under conjugation by K, while conjugation by Q permutes the displayed cross-commutators. By construction $K/N = (K^{\mathrm{ab}})_Q$, and Q acts trivially on this quotient. Hence

$$G/N \simeq (K^{\mathrm{ab}})_Q \times Q.$$

Since $N \subseteq [G, G]$, abelianizing the last display gives (5.21). □

The sentence in [3, Lemma 3.2, pp. 84–85] that passes from a split fundamental-group extension to a direct sum with $H_1(F;\mathbb{Z})$ is therefore valid only when monodromy acts trivially after abelianization. A minimal counterexample retaining a free quotient group is

$$G = \mathbb{Z} \rtimes_{-1} \mathbb{Z}.$$

The extension splits, but $G^{\mathrm{ab}} \simeq \mathbb{Z} \oplus \mathbb{Z}/2$, whereas the printed direct-product formula gives $\mathbb{Z}^2$. For the candidate's printed monodromy, theorem 5.3 gives

$$H^1(F;\mathbb{Q})^{\pi_1(B^\circ)} = \mathbb{Q}\gamma,$$

so the rational coinvariant rank is one, not $b_1(F) = 4$.

If C° is a sphere punctured at s points, the correct rank formula is

$$b_1(X^\circ) = s - 1 + \dim_{\mathbb{Q}} H_1(F;\mathbb{Q})_{\pi_1(C^\circ)}. \tag{5.22}$$

Thus the proof of CDP98 Theorem 3.1 reaches only the coinvariant rank unless an additional theorem proves trivial monodromy. CDP20 Lemma 4.2 repeats the direct-sum formula, and CDP20 Corollary 5.2 imports CDP98 Theorem 3.1. Repairing the local-freeness issue discussed there does not repair this distinct monodromy error. The torus branch of the CDP20 proof therefore has an unresolved dependency in addition to the singular-fibre defect above.

11.8 Exact present disposition

1. **[REFUTED]** The p. 692 reduction in [2] is not a formal consequence of the two cotangent sequences. Theorem 11.6 is a proper geometric counterexample to that exact inference.

2. **[REFUTED]** The direct-product homology formula in [3, Lemma 3.2] omits monodromy coinvariants. This leaves the torus branch imported by [2] without its printed rank argument.

3. **[VERIFIED]** The normalization–conductor construction, Cartier-fibre duality, flatness over the base curve, and top-degree analytic base change are proved above under their stated hypotheses.

4. **[VERIFIED]** The period-family descent, toric and finite fillings, global gluing, and integral analytic nearby-cycle/Leray bridges are proved in sections 6 and 7. Therefore the candidate's hypotheses (H1)–(H4) hold and $R^2 f_*(T_X \otimes L) \neq 0$ for every L by theorem 11.11.

5. **[VERIFIED]** Those same construction and topology results, together with theorem 8.4, hold for the constructed compact complex threefold. Hence $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, $a(X) = 1$, f is a holomorphic algebraic reduction, and $c_3(X) = 2$. The resulting X satisfies every stated hypothesis of CDP Theorem 2.2 and contradicts its conclusion; it is therefore a counterexample to that theorem as stated. This theorem-level conclusion uses the complete construction and topology proofs; it is not inferred from the two printed proof defects alone.

6. **[OPEN]** No independent repair of CDP Theorem 2.2 that bypasses both refuted dependencies is recorded here. Because the counterexample would be major, independent human verification of every candidate proof above remains essential; that social verification requirement is not disguised as a mathematical hypothesis.

Validation, repair, or disproof test 11.13 (decisive theorem-level outcomes)**.** The current workbench records two exact defects in the printed CDP chain and a completed route from the constructed threefold to a theorem-level counterexample. The remaining independent challenge is to falsify a named step in that route or find an independent repair of CDP Theorem 2.2 that proves one of the established candidate assertions impossible. Merely invoking the published theorem, without repairing its dependencies, does not identify such a step; neither would the two local printed-proof defects by themselves have decided the published theorem before the candidate was proved.

12 Confirmed local repairs kept separate from the theorem

These corrections are mathematically definite but do not settle the central construction.

1. **[VERIFIED]** Proposition 4.7(iv)'s closed-image conclusion is correct, although its printed justification is not. If $s : B \to E$ is the section of $f : E \to B$, then

$$s(B) = \{x \in E : s(f(x)) = x\},$$

the equalizer of id_E and $s \circ f$. Since E is Hausdorff, its diagonal is closed, so this equalizer is closed. Thus the explicit holomorphic section, its cusp disc, and the null-homotopy used in theorems 7.3 and 7.6 require no additional chartwise or local-finiteness hypothesis.

2. **[REFUTED]** Remark 7.28 is false for a fixed triple. For example,

$$(\ell_0, \ell_1, \ell_2) = (100, 1, -1), \qquad p = 12(100) - 4(1) - 3(-1) = 1199 \neq \pm 1.$$

The intended statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1, ℓ_2).

3. **[VERIFIED]** Appendix Lemma A.3 prints cohomological arrows while labelling them as homology. The sequence must be written

$$H^k(W) \longrightarrow H^k(\widetilde{W}) \oplus H^k(D) \longrightarrow H^k(\widetilde{D}) \longrightarrow H^{k+1}(W),$$

followed by universal coefficients when homology ranks are required.

4. **[VERIFIED]** Appendix Remark A.6 must read $\pi_1(N_0) = \bar{\Lambda} = \pi_1(W)$, not $\pi_1(N_0) = \Lambda$.

5. **[VERIFIED]** The modular assembler has a page-71 portability defect under the tested MiKTEX route; the canonical monolithic source renders correctly. This is a build defect, not a mathematical counterexample.

13 Tool and computational resource disclosure

The source manuscript available for this audit was reconstructed from a PDF in successive ten-page units in ChatGPT Professional-mode sessions and then consolidated into a single LaTeX source. Two separate ChatGPT Professional-mode sessions produced the first full-manuscript proof audits. The present consolidation, citation audit, exact-arithmetic reproduction, and validate/repair/disprove workbench were prepared with Codex using ChatGPT 5.6 Sol in Ultra mode. The exact computations reported here use integer and rational arithmetic; their mathematical certificates are reproduced in the body rather than delegated to software output.

These systems are disclosed as tools, not treated as mathematical authorities or authors. Every imported claim is checked against a publicly resolvable human source, and every new assertion remains subject to human verification. This disclosure and the point-of-use citation policy follow the recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [1]. Responsibility for any version submitted or published, including the adequacy of its proofs and references, belongs to the human author or editor who releases that version.

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